Highlights

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping of Stored Family Codes

Rodrigo Almeida de Oliveira, Juliano de Paulo Ribeiro, Edson Emilio Scalabrin

- Human check of the LLM screen: Recall 0.630 and 0.725.
- This revision retains 455 studies after two removal passes.
- F2 and F5 are withdrawn. F1, F3, and F4 are not findings.
- One stored code joins mining, stochastic models, and forecasting.
- Replication package: Zenodo DOI 10.5281/zenodo.21939600.

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping of Stored Family Codes

Rodrigo Almeida de Oliveira[a,1,*], Juliano de Paulo Ribeiro[a,2], Edson Emilio Scalabrin[a,3]

[a]*Pontifícia Universidade Católica do Paraná (PUCPR), Graduate Program in Informatics (PPGIa), Curitiba, PR, Brazil*

Abstract

Context. Software processes generate event data that process mining (PM) and stochastic models can use for forecasting. This mapping records family codes for that intersection on named cuts of one corpus. It does not establish how often the literature combines them for software development lifecycle (SDLC) prediction. **Objective.** To record the family codes a model stored on two cuts of this corpus, 150 and 292 studies, which were not re-marked, and to propose a research agenda. Those codes are not a census of the literature, because the screen missed eligible studies. **Method.** We report a systematic mapping study against SEGRESS [1]. The mapping protocol was not pre-registered. The human check of the screen was. Searches ran on 12 April 2026. An LLM screened 2,340 working-set rows and 3,807 auxiliary rows. Those lists share 364 papers under different identifiers. Their union is the 5,783 deduplicated records. The success bar, set before the judgements, was Recall ≥ 0.90. **Results.** The LLM list has 340 studies. Two predate 1994, leaving 338. Human Recall on the working-set sample is 0.630 (29 of 46; Wilson 95% CI 0.486–0.755). Auxiliary Recall is 0.725 (235 of 324). Removing 33 coded studies and keeping 150 added studies leaves a corpus of 455. F2 and F5 are withdrawn. F1, F3, and F4 were not re-marked and are not findings. One stored code in 379 matches process mining, a stochastic model, and forecasting. **Conclusion.** The agenda is not inferred from those codes. The mapping is not a census of the 455, because the screen missed eligible studies. PATHCAST is future work, not a result of this mapping. **Limitations.** The second pass used the abstract for almost every record. Seventy-six added studies were read from the title alone.

Keywords: Systematic Mapping Study, Process Mining, Stochastic Modeling, Software Process Forecasting, LLM-Assisted Screening, Mining Software Repositories

1. Introduction

Software organisations routinely need to answer forecasting questions about their own development process: when a feature will be delivered, how long an issue will remain open, or whether a release will slip. The data needed to

*Corresponding author.

Email addresses: rodrigo.almeida1@pucpr.edu.br (Rodrigo Almeida de Oliveira), juliano.ribeiro@pucpr.edu.br (Juliano de Paulo Ribeiro), scalabrin@ppgia.pucpr.br (Edson Emilio Scalabrin)

[1]ORCID: 0009-0009-6310-4126

[2]ORCID: 0009-0004-3605-485X

[3]ORCID: 0000-0002-3918-1799

answer them already exists as a by-product of everyday work—commits, issue transitions, pull requests, code reviews, and CI/CD executions are all time-stamped events. Three research communities have developed techniques that can exploit these traces, largely independently of one another. *Process mining* (PM) discovers, checks, and enhances process models from event logs, and has been applied to software repositories since the work of Cook and Wolf [2, 3] and Rubin et al. [4]. *Stochastic modelling*—Markov chains, stochastic Petri nets, Monte Carlo simulation—has a long tradition in software reliability and process simulation [5, 6]. *Machine-learning-based prediction* dominates mining-software-repositories research on defects, effort, and outcomes [7, 8], and, in the business-process domain, predictive process monitoring [9]. Combining them is attractive because each covers a weakness of the others: PM provides an empirically grounded process structure, stochastic models turn that structure into probability distributions over future paths and durations, and ML can correct systematic errors using context features. Whether, and how far, the literature has actually combined them for software processes is not known.

Related secondary studies.. Existing secondary studies each cover one side of this space. In business process management, predictive process monitoring has been surveyed and benchmarked for outcome prediction [10], remaining-time prediction [11], and in general [9], but on business event logs rather than software development processes. General mappings of process mining applications [12, 13] treat software engineering as one domain among many and do not address forecasting. Within software engineering, secondary studies have covered process simulation modelling [14, 6], process mining of FLOSS event logs [15], software process mining with Petri nets [16], continuous software monitoring backed by process mining [17], repository mining [7], and defect prediction [8]. Four of these ([14, 15, 16, 17]) fall within our scope and were retained as included studies under EC4 (Table 6), which excludes only secondary studies that do *not* address PM or stochastic methods in software engineering. None of them maps the joint use of process mining, stochastic modelling, and forecasting on software development data. This study records the family codes stored on the cuts it names. It does not establish how often the literature combines the three.

Why the mapping matters.. On the cuts of 150 and of 292, the model stored the family codes reported below, and those codes were not re-marked. The screen missed eligible studies, so the codes are not a rate for the literature. The SPMF taxonomy and the four agenda directions are proposals by the authors. For the methodology of secondary studies, the review is also a deployment of LLM-assisted screening whose selection error we measure against a pre-registered human reference standard, following LLM4SCREENLIT [18]. The mapping protocol itself was not pre-registered. The human check of the screen was pre-registered before the round-2 judgements.

Study design and contributions.. We report a systematic mapping study against SEGRESS [1], following the mapping guidelines of Petersen et al. [19], the evidence-based software engineering guidelines of Kitchenham and Charters [20], and the snowballing procedure of Wohlin [21]. The review addresses three research questions, each decomposed into two sub-questions. Table 1 names the output of each one: a stored family code on the cut named in the cell, or a field left unreported. Its contributions are (i) a classification of the coded rows, on the cuts named

in each sentence, by SDLC phase, data source, and technique family; (ii) stored family-code counts (F1, F3, F4), which this revision does not use as findings, and two withdrawn model counts (F2, F5); (iii) the SPMF taxonomy and a four-direction research agenda that is not inferred from F1–F5; and (iv) a measured account of the LLM-assisted selection, including where it failed. The authors' own integrated-pipeline proposal (PATHCAST) is mentioned only as one research-agenda hypothesis (Section 5.2) and is not evaluated here.

Section 2 describes the protocol; Section 3 the search and selection; Section 4 the results per research question; Section 5 the coded counts, taxonomy, and agenda; Section 6 threats to validity, including the human validation of the selection; and Section 7 concludes.

2. Review Protocol

2.1. Research Questions

The questions below name the cuts on which stored family codes are reported. Section 5.1 states once that F1, F3, and F4 are stored codes and do not motivate RA1–RA4. Later sections report the counts and do not repeat that status.

Table 1: Research questions and the output this revision reports. A stored code is the family mark on the cut named in the cell. "Not reported" means the field is free text and the exact match on 31 studies is 0%, or the field was not recomputed.

RQ	Sub-RQ	Question	Output in this revision
RQ1	RQ1.1	Which SDLC phases have been analyzed with process mining?	Not reported. The phase field was not recomputed.
RQ1	RQ1.2	What event log sources are used and how are they constructed?	Not reported. Free-text fields; exact match 0% on 31 studies.
RQ2	RQ2.1	Which process discovery algorithms have been applied to SDLC?	Not reported. Algorithm names are free text and were not re-extracted. Stored TF-PM count on the 150: 103.
RQ2	RQ2.2	Which predictive and stochastic techniques have been used?	Stored TF-ST codes on the 150 (49) and the 292 (152). Technique tokens are not a partition of TF-ST.
RQ3	RQ3.1	What is the level of integration between PM and stochastic methods?	Stored L0–L3 codes on the 150: 121, 28, 1, and 0.
RQ3	RQ3.2	What gaps exist for process-aware forecasting in the SDLC?	Not reported as a gap matrix. RA1–RA4 are proposals by the authors.

2.2. Search Strategy

The search strategy combines automated database search with backward and forward snowballing. The automated search targets five digital libraries: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and SpringerLink.

2.2.1. PICO Decomposition

The search string construction follows the PICO framework adapted for software engineering, as shown in Table 2.

Table 2: PICO decomposition for search string construction.

Facet	**Role**	**Concepts**
Population	Application domain	Software engineering, SDLC, software process, DevOps, CI/CD
Intervention	Analytical technique	Process mining, Markov chain, Monte Carlo, predictive PM, ML
Comparison	N/A	(omitted)
Outcome	Result	Forecasting, prediction, lead time, cycle time, process model

2.2.2. Search Strings

Three complementary search strings are executed to maximize coverage while maintaining precision.

Principal String ($P \wedge I$).. Block P captures the software engineering context:

```
("software engineering" OR "software development" OR "software process" OR "software lifecycle"
OR "software life cycle" OR "SDLC" OR "development lifecycle" OR "development life cycle"
OR "DevOps" OR "CI/CD" OR "continuous integration" OR "continuous delivery" OR "agile development"
OR "agile process" OR "issue tracking" OR "bug tracking" OR "version control" OR "code review"
OR "pull request" OR "software project" OR "software repository" OR "software maintenance"
OR "software evolution")
```

Block I captures the process mining and stochastic modeling techniques:

```
("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining" OR
"predictive process monitoring" OR "process prediction" OR "remaining time prediction" OR
"process forecasting" OR "event log" OR "process simulation" OR "Monte Carlo simulation" OR
"Markov chain" OR "Markov model" OR "stochastic process model" OR "transition probability"
OR "absorbing chain" OR "lead time prediction" OR "cycle time prediction")
```

Complementary String (MSR $\rightarrow$ Process Modeling).. Captures studies using MSR terminology without canonical process mining terms:

```
("mining software repositories" OR "software repository mining" OR "GitHub mining" OR "Jira
mining") AND ("process model" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "lead time"
OR "cycle time" OR "throughput" OR "transition matrix")
```

Frontier String (Stochastic ∧ PM ∧ SE).. Targets the intersection of stochastic methods with process mining in software contexts:

```
("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast" OR
"transition matrix") AND ("process mining" OR "event log" OR "workflow mining" OR "process
model" OR "business process") AND ("software development" OR "software engineering" OR "software
process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log" OR "build pipeline")
```

The blocks above are the shared term lists. They are not the Scopus syntax that was run. Table 3 is the ledger of every export in the corpus of 8,347 records: database, query name, and raw count. Every row was retrieved on 12 April 2026. The six Scopus rows sum to 2,377. All six Scopus strings, the four Web of Science strings, the two ACM query strings, and the four IEEE protocol strings are printed below. `scopus_principal` returned 1,595 records. None of the strings in the ledger contains a machine-learning block.

`scopus_principal` (1,595 records).. `TITLE-ABS-KEY(("software engineering" OR "software development" OR "software process" OR "software lifecycle" OR "SDLC" OR "DevOps" OR "CI/CD" OR "continuous integration" OR "agile development" OR "issue tracking" OR "version control" OR "code review" OR "pull request" OR "software repository" OR "software evolution") AND ("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining" OR "predictive process monitoring" OR "event log analysis" OR "process intelligence" OR ("Monte Carlo simulation" AND ("software" OR "process mining" OR "throughput" OR "lead time")) OR ("Markov chain" AND ("software" OR "process mining" OR "workflow" OR "development process")) OR (("stochastic model" OR "transition probability" OR "absorbing chain") AND ("software" OR "process mining")) OR (("remaining time prediction" OR "outcome prediction" OR "cycle time prediction" OR "lead time prediction") AND "software") OR ("event log" AND ("software" OR "repository" OR "JIRA" OR "GitHub")))) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")`

`scopus_markov_testing` (498 records).. `TITLE-ABS-KEY((("Markov chain" OR "Markov model") AND ("statistical software testing" OR "software testing" OR "software reliability")) OR ("A Markov chain model for statistical software testing")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")`

`scopus_stochastic_petri` (6 records).. `TITLE-ABS-KEY((("stochastic Petri net" OR "stochastic workflow net" OR "GSPN") AND ("remaining service time" OR "service execution time" OR "remaining time")) OR ("Prediction of Remaining Service Execution Time Using Stochastic Petri Nets with Arbitrary Firing Delays")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")`

`scopus_simulation_bridge` (200 records).. `TITLE-ABS-KEY((("process mining" OR "process discovery" OR "workflow mining") AND ("simulation model" OR "simulation models" OR "process simulation" OR "discovering simulation models" OR "colored Petri") AND ("event log" OR "process model" OR "business process")) OR ("Discovering Simulation Models")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")`

scopus_complementar (48 records).. TITLE-ABS-KEY(("mining software repositories" OR "software repository mining" OR "software archaeology" OR "mining commit logs" OR "GitHub mining" OR "Jira mining") AND ("process model" OR "workflow" OR "state transition" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "lead time" OR "cycle time" OR "throughput" OR "transition matrix" OR "sojourn time")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")

scopus_fronteira (30 records).. TITLE-ABS-KEY(("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast" OR "stochastic model" OR "absorbing Markov" OR "transition matrix") AND ("process mining" OR "event log" OR "workflow mining" OR "process model" OR "business process") AND ("software development" OR "software engineering" OR "software process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log" OR "build pipeline")) AND PUBYEAR > 1993 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE, "ar") OR LIMIT-TO(DOCTYPE, "cp")) AND LIMIT-TO(LANGUAGE, "English")

wos_principal (632 records).. TS=("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining" OR "event log analysis" OR "process intelligence" OR "predictive process monitoring") AND TS=("software engineering" OR "software development" OR "software process" OR "SDLC" OR "DevOps" OR "continuous integration" OR "agile development" OR "issue tracking" OR "code review" OR "software project" OR "software repository" OR "software maintenance" OR "software evolution")

wos_complementar (338 records).. TS=(("Monte Carlo simulation" OR "Markov chain" OR "stochastic model" OR "transition probability" OR "absorbing chain" OR "process simulation") AND ("software" OR "process mining" OR "workflow" OR "throughput")) AND TS=("software engineering" OR "software development" OR "software process" OR "SDLC" OR "DevOps")

wos_msr (26 records).. TS=("mining software repositories" OR "software repository mining" OR "software archaeology" OR "mining commit logs" OR "GitHub mining" OR "Jira mining") AND TS=("process model" OR "workflow" OR "state transition" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "lead time" OR "cycle time" OR "throughput" OR "transition matrix" OR "sojourn time")

wos_fronteira (1 record).. TS=("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast" OR "stochastic model" OR "absorbing Markov" OR "transition matrix") AND TS=("process mining" OR "event log" OR "workflow mining" OR "process model" OR "business process") AND TS=("software development" OR "software engineering" OR "software process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log" OR "build pipeline")

acm_principal (24 records).. (Abstract:"process mining" OR Abstract:"conformance checking" OR Abstract:"process discovery" OR Abstract:"workflow mining" OR Abstract:"event log analysis" OR Abstract:"process intelligence" OR Abstract:"predictive process monitoring") AND (Abstract:"software engineering" OR Abstract:"software development" OR Abstract:"software process" OR Abstract:"SDLC" OR Abstract:"DevOps" OR Abstract:"continuous integration" OR Abstract:"agile development" OR Abstract:"issue tracking" OR Abstract:"software repository" OR Abstract:"software project")

acm_msr (32 records).. (Abstract:"mining software repositories" OR Abstract:"software repository mining" OR Abstract:"software archaeology" OR Abstract:"GitHub mining" OR Abstract:"Jira mining") AND (Abstract:"process model" OR Abstract:"workflow" OR Abstract:"state transition" OR Abstract:"Markov" OR Abstract:"Monte Carlo" OR

Abstract:"lead time" OR Abstract:"cycle time" OR Abstract:"throughput" OR Abstract:"process mining" OR Abstract:"process analysis")

The third ACM export, `acm_extra_refs` (40 records), is a supplementary BibTEX file and has no query string. The four IEEE protocol strings are printed next. The raw file records 17 manual batches and 1,386 records. It does not record which of these four strings produced each batch. `acm_complementar` and `springer_11`–`springer_13` are in `config/queries.py` and were not executed.

ieee_exec1.. ("process mining" OR "conformance checking" OR "process discovery" OR "workflow mining" OR "event log analysis" OR "predictive process monitoring") AND ("software engineering" OR "software development" OR "software process" OR "SDLC" OR "DevOps" OR "continuous integration" OR "agile development")

ieee_exec2.. (("Monte Carlo simulation" OR "Markov chain" OR "stochastic process model" OR "process simulation") AND ("software" OR "process mining" OR "throughput" OR "lead time")) AND ("software engineering" OR "software development" OR "software process" OR "SDLC" OR "software testing")

ieee_exec3.. (("remaining time prediction" OR "lead time prediction" OR "cycle time prediction" OR "process forecasting" OR "outcome prediction" OR "event log construction") AND "software") AND ("software engineering" OR "software development" OR "software process")

ieee_exec4.. ("mining software repositories" OR "software repository mining" OR "software archaeology" OR "mining commit logs" OR "GitHub mining" OR "Jira mining") AND ("process model" OR "workflow" OR "state transition" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "process mining" OR "process analysis")

The SpringerLink cell in the ledger is the string that was run.

The eligibility window (IC1) runs from 1 January 1994 to the search execution date; publications dated 2026 are therefore partial-year coverage (indexed by 12 April 2026) and are reported as such. On 31 October 2026 a fourth string, with `machine learning`, `deep learning`, `neural network`, `random forest`, `gradient boosting`, `xgboost`, or `lstm`, and with lead time, cycle time, remaining time, throughput, or just-in-time defect as the target, was run on Scopus. It returned 165 records. Title screening against the criteria in Table 6 left 19 includes, all under IC4c, and 2 records already in the 455. None of the 19 joins process mining, a stochastic model, and forecasting. They are not added to the 455: the screen used the title only, and the other four databases were not re-run. F5 is withdrawn, so those 19 are not used as a finding. Where a database supported a date restriction, the query used publication years 1994–2026; the Web of Science export was not date-restricted at query time, and IC1 is enforced at screening for every source (Table 4, note).

2.2.3. Snowballing Procedure

Snowballing follows Wohlin [21] for one iteration. The identification file stores 595 records, each labeled as coming from both the reference list and the citation list of one source DOI. Those labels name 400 distinct source DOIs. The file does not split the 595 into a backward count and a forward count, and it does not record a second iteration. The stop rule is that single pass: a candidate is kept when the title or the abstract contains one process-mining or stochastic term and one software-engineering term. The file does not name an evaluator, does not give a

Table 3: Search ledger. Each row is one export retrieved on 12 April 2026. The rows sum to 8,347.

Database	**Query**	**String**	**Raw**
Scopus	scopus_principal	printed above	1,595
Scopus	scopus_complementar	printed above	48
Scopus	scopus_markov_testing	printed above	498
Scopus	scopus_stochastic_petri	printed above	6
Scopus	scopus_simulation_bridge	printed above	200
Scopus	scopus_fronteira	printed above	30
Web of Science	wos_principal	printed above	632
Web of Science	wos_complementar	printed above	338
Web of Science	wos_msr	printed above	26
Web of Science	wos_fronteira	printed above	1
ACM DL	acm_principal	printed above	24
ACM DL	acm_msr	printed above	32
ACM DL	acm_extra_refs	supplementary BibTEX export	40
IEEE Xplore	ieee_manual_01–17	17 batches; strings printed above	1,386
SpringerLink	springer_1	`"process mining" "software development"`	527
SpringerLink	springer_2	`"process mining" "software process"`	167
SpringerLink	springer_3	`"process mining" "software engineering"`	743
SpringerLink	springer_4	`"Markov chain" "software process"`	743
SpringerLink	springer_5	`"Monte Carlo" "software process"`	70
SpringerLink	springer_6	`"mining software repositories" "process model"`	140
SpringerLink	springer_7	`"stochastic" "software process" "event log"`	46
SpringerLink	springer_8	`"predictive process monitoring" "software"`	16
SpringerLink	springer_9	`"conformance checking" "software development"`	132
SpringerLink	springer_10	`"workflow mining" "software"`	300
Snowballing	snowball_both	one iteration; labels do not split direction	595
Control set	control_papers_bib	validation only	12
Total			**8,347**

Table 4: Database-specific search adaptations and raw retrieval counts.

Database	Search Field	Strings	Filters at query time	Records
Scopus	TITLE-ABS-KEY	6	ar, cp; EN; PUBYEAR 1994–2026	2,377
IEEE Xplore	All Metadata	17 batches	Conf. + Journals; 1994–2026	1,386
Web of Science	TS (Topic)	4	Article, Proc.; EN; no date filter	997
SpringerLink	Simplified	10 queries	Chapter, Article; EN	2,884
ACM DL	Abstract	3	Date: 1994–2026	96
Snowballing	–	–	Backward + forward	595
Control set	–	–	Manual	12
Total				**8,347**

Note: all searches executed on 12 April 2026. After deduplication, 7 records dated before 1994 remained in the pool (6 from Web of Science, 1 from ACM DL). IC1 (1 January 1994 to the execution date) is enforced at screening; two pre-1994 auxiliary-tier records that had been confirmed in R1 were removed in this revision (Appendix A).

count per source DOI, does not record how duplicates were removed inside the iteration, and was not restarted from the studies later marked as included. The ten papers in Table 5 check whether the database strings retrieve known studies. They are not the 400 source DOIs. The database searches are dated 12 April 2026. The snowball file does not carry a separate date.

2.2.4. Search String Validation

The search strings are validated against a control set of 10 papers that are known to be relevant to the review scope (Table 5). The ten papers were known when the strings were written. Retrieving all ten shows that the strings can return these known papers. It does not measure how many relevant studies the strings miss, and it is not recorded as a pass.

2.3. Inclusion and Exclusion Criteria

The inclusion criteria are organized in two tiers. Formal criteria (IC1–IC3) filter on publication attributes and are applied during the database search configuration. Content criteria (IC4a–IC4d) require that the study's research contribution addresses at least one of four specific technique-domain intersections; they are applied during title/abstract screening and during the second-pass eligibility screen. The replication files label that second pass "FT". For almost every record the second pass read the title and the abstract. "Full text" in this article means a PDF was read. Exclusion criteria (EC1–EC5) are applied cumulatively across both screening phases. Criteria EC1–EC4 are evaluated by the LLM-assisted screener; EC5 (full text inaccessible) is applied programmatically after the inter-library-loan recovery pass for Band D/E papers fails to retrieve a copy. In the auxiliary tier, EC5 is extended to cover non-recoverable metadata insufficiency: papers for which no usable abstract could be recovered after the multi-source enrichment

Table 5: Control papers for search string validation.

#	Paper	Justification	Captured via
V1	Cook & Wolf (1998)	Seminal PM + SE	ACM
V2	Rubin et al. (2007)	PM framework for SW processes	Scopus
V3	Poncin et al. (2011)	PM + MSR intersection	IEEE
V4	Caldeira & Cardoso (2019)	PM + SW process assessment	IEEE
V5	Lemos et al. (2011)	PM for SW process management	IEEE
V6	Whittaker & Thomason (1994)	Markov + SW testing (seminal)	Scopus
V7	Tax et al. (2017)	PPM seminal (BPM baseline)	Control
V8	Rogge-Solti & Weske (2015)	Stochastic + PM forecasting	Scopus
V9	Rozinat et al. (2009)	PM → Simulation bridge	Control
V10	Gupta & Sureka (2017)	PM multi-repository in SE	ACM

cascade are flagged for human review in the revised protocol. In the **original execution run**, papers for which the primary and verifier raters produced an irreconcilable decision without accessible full text were closed under EC5 rather than escalated to human arbitration; Section 6 quantifies the resulting Lost Evidence against the pre-registered human check. That check is a two-rater sample. It is not a double screen of every record. The revised protocol treats T/A `maybe` and FT `pending` as positive referrals (LLM4SCREENLIT [18] R6).

At least one of IC4a–IC4d must be satisfied for inclusion. Criterion IC4b ensures that seminal works applying Markov chains to software processes—such as Whittaker and Thomason [5]—are captured despite predating the term "process mining." To avoid colliding with the publication filters IC1–IC3, Results and Appendix A report the four content families as **TF-PM**, **TF-ST**, **TF-FC**, and **TF-MSR** (IC4a–IC4d, respectively).

2.4. LLM-Assisted Screening Methodology

Given the scale of the working set (2,340 papers) and the requirement for systematic, reproducible screening decisions with traceable rationale, the title/abstract screening phase employs a language model as the primary screener. Specifically, *claude-haiku-4-5-20251001* is used via the Anthropic Message Batches API. The screening, QA, extraction, and verifier calls pass the model identifier and `max_tokens` and do not pass a temperature. The Messages API then uses its default temperature of 1. The prompt encodes the IC4a–IC4d and EC1–EC4 criteria (EC5 is applied programmatically post-screening), a short scope statement of the target intersection, and decision rules for the three-class output: `include`, `exclude`, or `maybe`. The scope statement was labelled with the name of the authors' research programme (PATHCAST); its content restates the four content families IC4a–IC4d and adds no further criterion, but

Table 6: Inclusion and exclusion criteria.

Type	ID	Criterion
Inclusion	IC1	Published between 1 January 1994 (from Whittaker & Thomason onward) and the search execution date (12 April 2026).
	IC2	Language: English.
	IC3	Peer-reviewed journal article, conference paper, or workshop paper (≥4 pages).
	IC4a	Applies process mining techniques (discovery, conformance checking, predictive process monitoring) to software development artifacts (commits, issues, pull requests, CI/CD logs, bug trackers, VCS).
	IC4b	Uses stochastic models (Markov chains, Monte Carlo, stochastic Petri nets, absorbing chains) to analyze software development processes or workflows.
	IC4c	Proposes or evaluates forecasting of software process metrics (lead time, cycle time, remaining time, throughput, defect rates) from event logs or repository data.
	IC4d	Mines software repositories (GitHub, Jira, VCS, CI/CD platforms) to discover, analyze, or improve software process models.
Exclusion	EC1	Application domain is exclusively outside software development (healthcare, manufacturing, supply chain) without demonstrated relevance to SDLC processes.
	EC2	Software appears only as the implementation platform; the process being analyzed is not a software development process.
	EC3	Purely algorithmic or theoretical contribution without empirical evaluation in a software development context.
	EC4	Secondary study (SLR, mapping study, survey) not specifically addressing PM or stochastic methods in software engineering.
	EC5	Full text or decision-critical metadata inaccessible despite all retrieval attempts (open access, institutional, ILL, author-provided preprint, multi-source abstract enrichment). Applied programmatically after Band D/E recovery and auxiliary enrichment; also closes irreconcilable ambiguous cases when no accessible full text exists.

we report it because a framework label in the prompt is a potential source of selection bias (Section 6). The exact prompts are in the replication package.

The screening prompt instructs the model to (i) identify which ICs are satisfied, (ii) identify which ECs apply, (iii) assign a decision with a one-to-two sentence rationale, and (iv) return the result as structured JSON. The `maybe` class captures genuine uncertainty—typically arising from missing abstracts or ambiguous domain terminology—without forcing a binary decision.

At the second-pass eligibility screen, the same model is applied with a stricter prompt that enforces a binary outcome (`include` or `exclude`) with `pending` reserved only when the abstract is genuinely insufficient for a decision. The FT prompt also instructed the model to prefer `exclude` when undecided between `include` and `exclude`; this rule biases the FT stage toward false negatives, which is the error the human validation in Section 6 is designed to measure.[4]

This approach extends prior work on LLM-assisted systematic reviews [22, 23, 18] to the software process mining domain. The primary screener produces structured outputs that enable post-hoc quality analysis. Validation follows a three-stage protocol: (i) a pre-registered two-rater check of a stratified random 20% sample, blind to the LLM (n = 468 at title/abstract, n = 177 at the second pass), which is the source of the Recall figures (Tables 17 and 18); (ii) cross-model verification on a separate stratified 20% sample, with *claude-sonnet-4-6* as second rater and Cohen's κ reported following Landis and Koch (1977) [24]; and (iii) a two-reviewer re-screen of the working-set records the LLM had excluded and of the auxiliary records that were pending or otherwise unclassified. On that re-screen the two reviewers agreed with 92% of the suggested labels, including the labels that a heuristic pass had proposed for the later S1 batches and for all of S3. The sheets for that re-screen (4,005 rows) store a decision equal to the suggestion on every row, so the labels they did not accept are not marked there. Rebuilt from the title or abstract the LLM treated as sufficient, that share of the coded includes is 33 of 338 (9.8%), listed in `labels_ta_insufficient_for_include.csv` and removed in Section 4. Cross-model verification is a secondary signal. The Recall figures come from the sample, not from the 92% agreement.

Operationally, the **original execution run** did not preserve an open-ended manual-arbitration queue: unresolved `maybe`/`pending` cases were closed under EC5-extended. The **revised protocol** routes such cases to human review instead. In the working set, T/A `maybe` decisions were forwarded to FT screening, so no human-relevant record in the T/A validation sample was lost at T/A (Section 6).

[4]The two protocols use different nomenclature for the uncertainty class: T/A uses `maybe` (epistemically broad: criterion ambiguity *or* abstract insufficiency), whereas FT uses `pending` (narrowly: abstract insufficiency only). The semantic difference is intentional — the T/A pass operates on metadata where ambiguity is expected, while the FT pass commits to binary decisions whenever the abstract permits and escalates only the abstract-insufficient subset to manual full-text retrieval. This mapping is preserved in the cross-model κ analysis (Tables 19 and 20).

2.5. Reporting-Quality Assessment

Under SEGRESS [1] §4.3.2, risk-of-bias assessment is optional for mapping studies; we retained a **reporting-quality** checklist (not a risk-of-bias instrument) adapted from Dybå and Dingsøyr [25], with eight binary criteria. It is *not* a selection filter, and the 4/8 line is not applied to any finding. The same model that screens and extracts (*claude-haiku-4-5-20251001*) also assigns the eight scores. A score produced from an abstract is an abstract-only score. It is reported as such and is not pooled with a score produced from a PDF. The checklist is descriptive. It is not a validity judgement on primary-study outcomes:

Table 7: Quality assessment checklist.

#	Criterion	Score
QA1	Are the research objectives clearly stated?	0/1
QA2	Is the software engineering context described in detail?	0/1
QA3	Is the data source (event log) described reproducibly?	0/1
QA4	Is the PM or stochastic technique formally defined?	0/1
QA5	Are results validated empirically?	0/1
QA6	Are threats to validity discussed?	0/1
QA7	Are data and/or code publicly released?	0/1
QA8	Are process model quality metrics reported (fitness, precision)?	0/1

The eight scores are descriptive. The same model that screens and extracts assigns them. No finding uses a cutoff on the sum.

QA scoring is performed by the same LLM-assisted protocol used for screening (Section 2.4): *claude-haiku-4-5-20251001* receives a strict rubric for each criterion and returns a binary score with a brief justification. The full prompt and the per-study scores are part of the replication package (Section 6.5). Table 8 reports the scores for the 150 working-set studies that remain.

The model scored 27 of the 150 working-set studies that remain below 4/8. The score is not a filter. Section 5.1 does not apply a cut on 4/8.

2.6. Data Extraction

For each included study, the attributes in Table 9 are extracted.

3. Search Results and Study Selection

3.1. Database Retrieval and Deduplication

The database rows in Table 3, apart from snowballing and the control set, returned 7,740 records. One snowballing iteration, stored as 595 records attributed to 400 source DOIs, yielded the additional records (Section 2.2).

Table 8: Quality scores for the 150 working-set studies that remain. The scores are descriptive and are not a filter.

Score band	**Count**	**Notes**
Studies assessed	150	Working-set studies still in the corpus. Not the corpus of 455.
Studies with score $\geq 4/8$	123	Score assigned by the model. Not a filter.
Studies with score $< 4/8$	27	Not a filter.
Mean QA score	4.90	Mean and standard deviation (1.36)
Median QA score	5.00	Median and interquartile range (4.00–6.00)

Table 9: Data extraction form.

Field	**Description**
ID	Unique identifier
Authors, Year, Venue	Bibliographic metadata
SDLC Phase(s)	Lifecycle phases covered
Event Log Source	Jira, GitHub, GitLab, Jenkins, etc.
Event Log Construction	Manual, semi-automatic, or automatic
PM Technique Category	Discovery / Conformance / Enhancement / Prediction
Specific Algorithm(s)	Alpha, Heuristics, Inductive Miner, etc.
Stochastic Method	Markov / Monte Carlo / Stochastic Petri Net / None
ML Technique	RF, XGBoost, LSTM, etc. / None
Prediction Target	Remaining time, outcome, lead time, delay, etc.
Integration Level	L0–L3 (derived from technique families; Table 15)
Validation Type	Case study / Experiment / Industrial / Simulation
Tool/Platform	PM4Py, ProM, Disco, custom, etc.
Dataset Size	Number of events and cases analyzed
Process Model Quality	Fitness, precision, generalization (if reported)

Together with 12 control papers maintained as a validation-only set, the initial corpus totaled 8,347 records. Table 10 summarizes the retrieval by source.

Table 10: Record retrieval by source.

Source	Raw Records	After Internal Dedup
Scopus (6 queries)	2,377	1,852
SpringerLink (10 queries)	2,884	1,543
IEEE Xplore (manual)	1,386	1,263
Snowballing	595	575
Web of Science (4 queries)	997	514
ACM Digital Library (3 queries)	96	34
Control set (validation only)	12	2
Total	**8,347**	**5,783**

Cross-source deduplication removed 2,564 duplicate records (30.7%), yielding 5,781 unique primary-corpus papers plus 2 unique control papers. Deduplication used normalized DOI matching, followed by fuzzy title matching (Levenshtein similarity ≥ 0.9) for records without DOIs.

3.2. Operational Working Set Construction

Screening 5,783 records in a single pass would require substantial resources. A working set of 2,340 papers was therefore constructed from the highest-signal primary queries—those whose query design directly targets the review intersection (process mining, stochastic modeling, and software development)—as shown in Table 11. Of the 2,340 working-set identifiers, 1,976 are among the 5,783 deduplicated records, the other 364 rows are the same papers under a different identifier inside the 3,807 (5,783 – 1,976), the union of the two identifier lists is the 5,783, and 345 of those 364 are Scopus rows in the working-set file, so 2,197 – 345 = 1,852, the Scopus count in the deduplicated file. Of the 364 pairs, 69 appear in the corpus of 455 under the working-set identifier and 30 under the canonical identifier. No pair appears under both identifiers. The 3,807 is the auxiliary identifier list that was screened.

Of the ten known control papers, eight are retrieved by the primary queries and all ten appear once a validation-only set is counted. That count does not measure search recall.

3.3. Title/Abstract Screening

All 2,340 working-set papers were screened against IC4a–IC4d and EC1–EC4 using the LLM-assisted protocol described in Section 2.4. The screening was executed across five sequential batches of up to 500 papers each via the Anthropic Batches API, producing structured JSON outputs with decision, rationale, and matched criteria for each record.

Table 11: Working set composition.

Database	Queries included	Rows
Scopus	scopus_principal, scopus_complementar, scopus_fronteira, scopus_markov_testing, scopus_stochastic_petri, scopus_simulation_bridge	2,197
IEEE Xplore	ieee_manual_01 (PM + SE subgroup)	97
ACM DL	acm_principal, acm_msr	46
Total		**2,340**

Table 12: Title/abstract screening results.

Decision	Count	%
Include	111	4.7
Maybe	775	33.1
Exclude	1,454	62.2
Total	**2,340**	100.0

The `include` set (111 papers) and the `maybe` set (775 papers) together form the second-pass queue of 886 papers. The `exclude` decisions were driven primarily by EC1 (domain outside software development, principally generic business process management) and EC3 (theoretical algorithm proposals without SE evaluation).

Information basis at T/A.. For 1,695 of the 2,340 working-set records (72.4%; mostly Scopus and ACM) the T/A decision was made on the title alone: the Scopus Search API leaves the abstract field empty for most records, and the separate Abstract Retrieval call was not part of the original extractor. This is one reason for the large `maybe` class. After the R1 review we recovered abstracts for 1,131 of these records (working-set abstract coverage 27.6% → 74.0%) and re-screened them with the same model, prompt, and decision rule; after discarding 44 recoveries that had attached the wrong abstract, 1,087 re-screenings remain, of which 372 (34.2%) changed decision and 4 moved from `exclude` to `include`/`maybe`. Those four records were not added as a separate batch. Studies that entered the corpus because a human later judged them eligible are the 168 described in Section 4, of which 18 are removed in the eligibility pass. The 150 that remain are the corpus added in this revision.

3.4. Second-Pass Eligibility Screening ("FT")

The 886 papers forwarded from the T/A phase entered the second-pass eligibility screen, using the strict prompt described in Section 2.4 (decisions limited to `include`, `exclude`, or `pending`; `maybe` is not permitted at this stage). To maximise the number of papers assessable by the LLM, a multi-source abstract-enrichment cascade was executed

prior to screening—combining DOI-based lookups via Semantic Scholar, OpenAlex, CORE, and Crossref, followed by title-based fallback lookups—raising abstract coverage from 16.6% to 72.6% (643 of 886 papers).[5]

Information basis at the second pass.. Despite its label, this stage was not a full-text reading for almost all records: the screening prompt received the same (enriched) title and abstract as T/A, under the stricter decision rule described in Section 2.4, and a PDF was read for only 0.5% of the 886 records. We keep the label "FT" for traceability with the replication package, but its decisions are abstract-based eligibility decisions. Full texts were read by humans only in the validation samples (Section 6) and, after R1, for 8 records with conflicting abstract-based decisions. Three of those eight were eligible on the PDF: `ba2ff831`, `ee562777`, and `9188583f`. The first two are already in the original confirmed list. The third is not in the 455-study corpus of this revision.

Table 13 summarises the outcome.

Table 13: Second-pass eligibility outcomes (886 papers forwarded from the T/A phase). The replication files label this stage FT. The prompt received the title and the abstract.

Outcome	**Count**	**%**
Included	169	19.1
Excluded	717	80.9
EC3 (algorithmic, no SE eval.)	(232)	(26.2)
EC1 (non-SE domain)	(205)	(23.1)
EC5 (full text inaccessible, post-audit active stratum)	(145)	(16.4)
EC2 (SW as implementation)	(110)	(12.4)
EC4 (out-of-scope secondary)	(60)	(6.8)
Total	**886**	100.0

Note: EC counts are non-exclusive — a single paper may trigger multiple ECs when more than one applies (e.g., a non-SE-domain paper whose contribution is purely theoretical). The EC5 count of 145 is the active post-audit stratum after re-screening the original historical EC5 band of 148 papers and reclassifying 3 of them to EC3. The sum of per-EC counts is 232 + 205 + 145 + 110 + 60 = 752, which exceeds the 717 excluded papers by 35 (corresponding to papers triggering more than one EC). The percentages above are with respect to the 886-paper FT queue.

Among the 717 excluded papers, EC3 (algorithmic or theoretical contribution without SE evaluation, $n = 232$) and EC1 (domain outside software development, $n = 205$) jointly account for the largest share of FT-stage exclusions. EC5 (full text inaccessible despite all retrieval attempts, active post-audit $n = 145$) corresponds to the residual portion of the original 148-paper Band D/E stratum for which no open-access copy, institutional access, or author-provided preprint could be located; these are recorded in a dedicated traceability file and documented in the PRISMA flow as a

[5]Papers were processed in order of a five-band priority scheme: Band A (91 papers, T/A `include` ≥2 ICs), Band B (20, T/A `include` 1 IC), Band C (39, T/A `maybe` + abstract), Band D (152, T/A `maybe` + IC signal, no abstract), Band E (584, T/A `maybe`, no IC signal, no abstract). Abstract enrichment and LLM re-screening were applied to Bands A, B, C, and the subset of D/E that gained an abstract after enrichment.

named exclusion stratum. EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary studies, $n = 60$) account for the remainder.

Figure 1 presents the complete PRISMA 2020 flow diagram [26] of the selection process.

4. Results and Synthesis

4.1. Overview of Included Studies

This revision includes **455 studies**: 305 that remain from the coded list of 338, plus 150 that remain after the eligibility pass.

The 338 are the LLM list of 340 studies, minus two papers from 1991 and 1992. The human re-screen marked 188 records as eligible. Twenty of them repeated a study already in the list, leaving 168. Eighteen of those 168 fail an inclusion or exclusion rule and are removed. Seventeen had been read from an abstract or a PDF: the extraction either names no technique in IC4a–IC4d, or the text is outside software development (EC1 or EC2). One, identifier `46815264`, is dated 1992 and fails IC1. Forty-six others had been labelled "none of the four" even though the extraction names a technique; they stay, with that family. The 18 removals and the 46 assignments are listed in `corpus_eligibility_pass.csv`. Of the 150 that remain, 11 were read as a PDF, 63 from the abstract, and 76 from the title only.

Thirty-three studies, 33 of the 338 coded includes (9.8%), are then removed. The re-screen sheets do not mark them: on all 4,005 rows the decision equals the suggestion. The list was rebuilt from the title or abstract the LLM treated as sufficient. Each was an include for which that text does not support inclusion. The object is generic process mining or forecasting, a business process, product execution, or system architecture. Two of them, `96767878` and `f449ac41`, have no abstract, and the inclusion note itself asks for the full text. Seventeen come from the working set, seven from the first auxiliary tier, and nine from the second. The list is `labels_ta_insufficient_for_include.csv`. The coded studies that remain are 305, and the corpus is 305 + 150 = 455.

The three technique families were marked on **379** of the 455: the 305, plus the 74 added studies that had an abstract or a PDF and meet at least one content criterion. One of those 379, PRIMAD, has all three families. The 76 studies known only by the title were left blank. A blank is not counted as a no.

Table 14 is the single count of every cut used in the paper. Four of those rows answer different questions, and they are not interchangeable. The context set is the 455. The 33 removed from the coded list are the LLM includes whose title or abstract does not support inclusion. The 74 added studies stay because the extraction names a technique in IC4a–IC4d. The 76 title-only studies were not marked, and a blank is not treated as satisfying a content criterion. The rows with technique-family codes filled before this round were the 338; 305 remain after the removal of 33. The rows with at least one family mark, including the 74 added studies, are the 379. The intersection of process mining, a stochastic model, and forecasting is one study inside that 379. Each research question is answered on the cut named in its sentence. The working-set family codes use 150 studies. That is the 169 extraction minus 17 studies whose title or

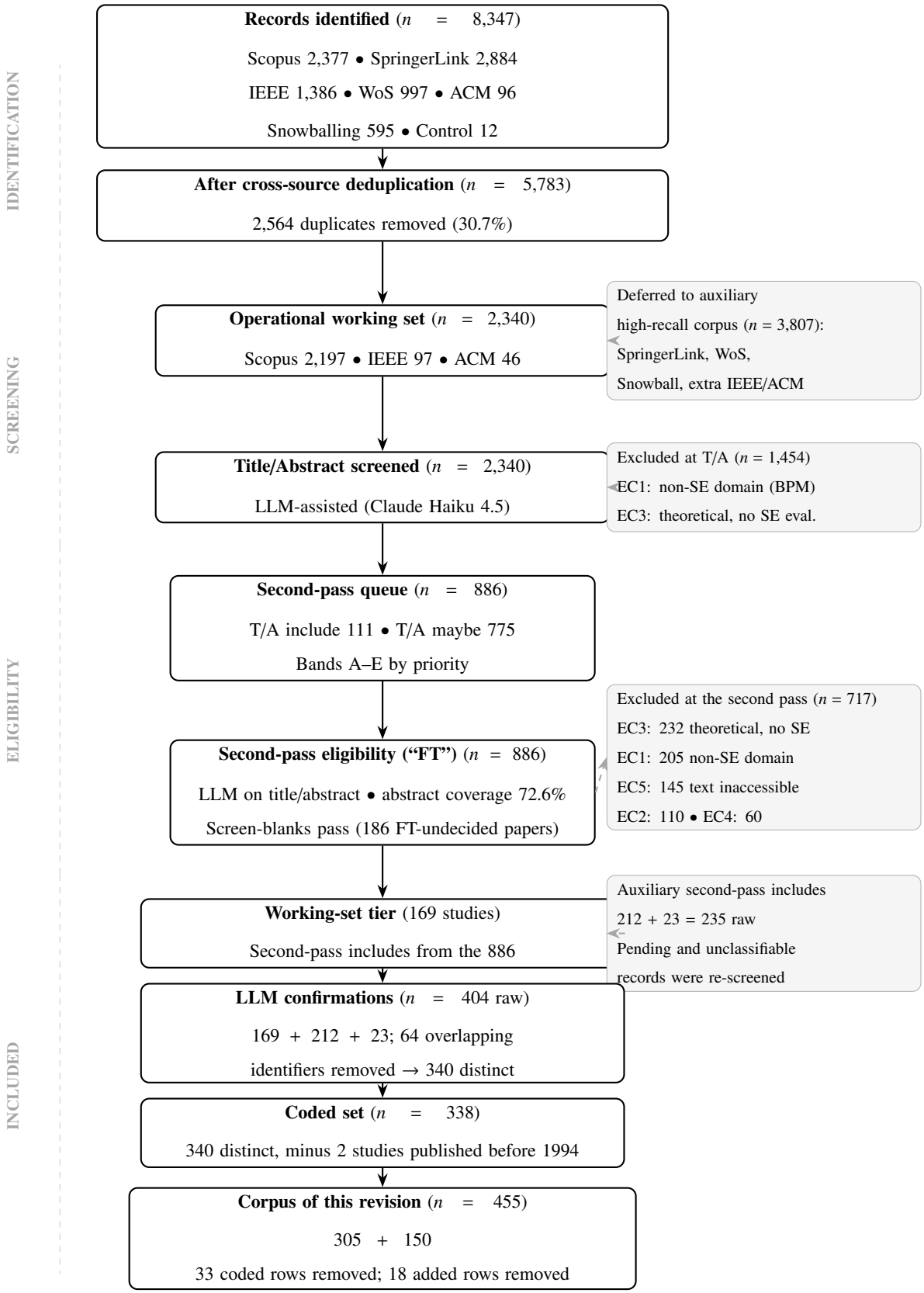


Figure 1: PRISMA 2020 flow of the study selection. The 2,340 and the 3,807 are identifier lists. They share 364 papers, and their union is the 5,783. The included phase continues through the auxiliary screen, the removal of overlapping identifiers, the removal of two pre-1994 studies, and the human re-screen. The mapped set is 455 studies (Table 14). The 318-study joint tables are an older cut (381 records minus 63 duplicates) and are not a node of this flow. In the original run, 595 records with no abstract and 16 records that stayed pending were closed under EC5; this revision re-screened those records.

abstract does not support inclusion, and minus two extra identifiers of one paper the corpus keeps once (`2647ea07`). The joint family codes use 292 studies. That is the 318 minus those 17, minus seven studies from the first auxiliary tier removed for the same reason, and minus the two pre-1994 papers. Nine of the 33 removed studies are not in the 318. The joint tables use 318. That 318 is what remained after 63 duplicate records were removed from an earlier combined list of 381. The 340 is what remained after 64 duplicate records were removed from 404 raw confirmations. The two cuts were built by different removals. Adding the 23 studies of the second auxiliary pass to 318 does not produce 340. The 33 removed above are the coded includes whose title or abstract does not support inclusion. The appendix lists the 455 studies and groups them by family.

Table 14: Counts used in this paper. Each row is a different cut.

Studies	**Meaning**
8,347	Records retrieved, before deduplication.
5,783	Records after deduplication.
2,340	Working-set identifier list. 1,976 of these identifiers are in the 5,783.
3,807	The other identifiers in the 5,783 (5,783 − 1,976). This list holds the second identifier of 364 papers that are also in the working set. The union is 5,783.
404	Raw LLM confirmations (169 + 212 + 23).
340	Distinct LLM list (404 minus 64 duplicate records).
338	The 340 minus two papers from 1991 and 1992.
381	Earlier combined list, before the 318 cut.
318	Joint tables (381 minus 63 duplicate records). Not 340 minus 23.
150	Working-set family codes. The 169 minus 17 removed studies and minus 2 extra identifiers of one paper.
292	Joint family codes. The 318 minus 24 removed studies and minus 2 pre-1994 papers.
33	Coded includes removed (33 of 338, 9.8%). The title or abstract does not support inclusion. Not marked on the 4,005-row sheets.
305	Coded studies that remain (338 minus 33).
455	This revision (305 + 150). Of 168 after de-duplication, 18 were also removed.
379	At least one family mark (305 + 74). Not a rate over the 455.
76	Added studies known only by the title. Marks left blank.

The list is `included_studies_506.csv` (455 rows; the filename keeps an earlier count). The file of 340 rows is

kept so the removal of the two pre-1994 papers can be checked. Thirty-five of the 150 added studies were assigned a model score of at least 4/8. A score taken from an abstract is not a quality assessment of the study, and no score is used to keep or drop a study. F2 and F5 are withdrawn, and their percentages are not reported. The percentages under F1, F3, and F4, apart from the count of one study in 379, are the family codes on the 150 and the 292. On the original 169, TF-PM with neither TF-ST nor TF-FC is 105, not 68: the 68 are the code that is exactly TF-PM, and 37 more are TF-PM with TF-MSR only. On the original 318 the same definition is 138, not 78: the 78 are TF-ST together with TF-FC and no TF-PM. The figures below use the 150 and the 292. Their status is stated once in Section 5.1.

The 150 working-set studies that remain span 1995–2026 (publications dated 2026 that were already indexed on 12 April 2026 are retained), with a clear acceleration from 2014 onward (Figure 2). One hundred and four of them (69.3%) appeared in 2015–2026. A completed SEGRESS [1] reporting checklist is in Appendix Appendix B.

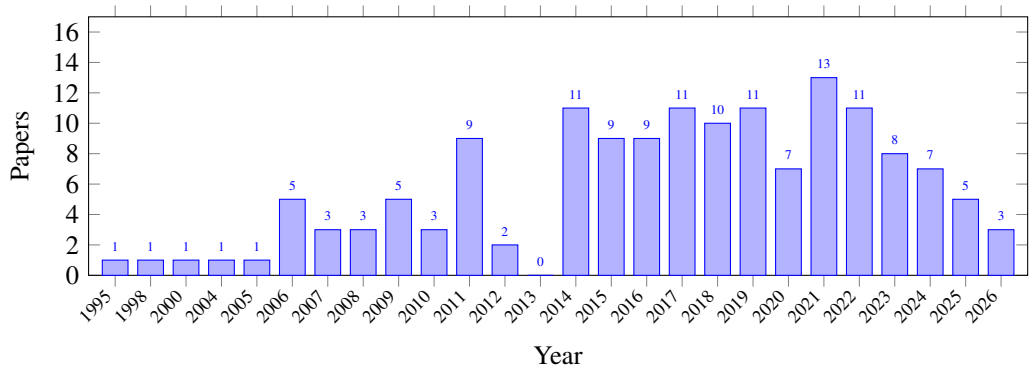


Figure 2: Publication year of the 150 working-set studies that remain.

By document type, the majority are conference papers and journal articles. By source database, Scopus is the dominant contributor, followed by IEEE Xplore and ACM DL. Data extraction was completed for all 169 studies using a structured LLM-assisted pipeline (*claude-haiku-4-5-20251001*) operating on locally available PDFs (75/169, of which 73 were operationally coded in the triangulation subset) or abstract/title metadata (94/169).

4.2. Inclusion Criterion Coverage

Figure 3 shows the distribution of inclusion criteria among the 150 working-set studies that remain (criteria are non-exclusive; a paper may satisfy multiple).

On the cut of 150, the model stored TF-PM on 103 rows (68.7%). That count was not re-marked and is not a rate for the literature. On the same cut the model stored TF-MSR on 42 rows (28.0%), TF-ST on 49 (32.7%), and TF-FC on 27 (18.0%). TF-PM with neither TF-ST nor TF-FC is 91 rows (60.7%). The other 101 of the 150 do not carry TF-ST, and 49 + 101 = 150. That split is the stored family-code count labeled F1 in Section 5.1.

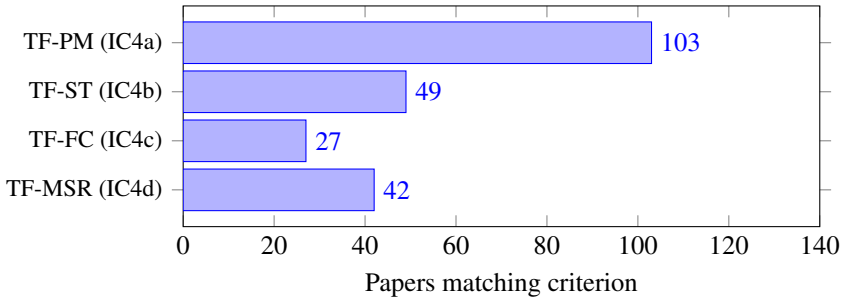


Figure 3: Inclusion criteria matched by the 150 working-set studies that remain. Counts are non-exclusive. TF-PM, TF-ST, TF-FC, and TF-MSR are the content criteria IC4a–IC4d. TF-PM with neither TF-ST nor TF-FC is 91 of these 150.

4.3. Prevalent Exclusion Patterns

Among the 717 FT-excluded papers (Table 13), the largest exclusion reason is EC3 (algorithmic or theoretical contribution without software-engineering evaluation, $n = 232$), indicating a substantial body of process mining and stochastic modeling research that has not been empirically applied to software engineering contexts. EC1 (domain exclusively outside software development, $n = 205$) is the second-largest reason and consists primarily of generic business process management (BPM) studies that use the same terminology as the search strings but apply techniques to organizational processes unrelated to SDLC. EC5 (full text inaccessible, active post-audit $n = 145$; original historical stratum $n = 148$) closes the group of large strata, followed by EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary study, $n = 60$). EC counts are non-exclusive: the sum of per-EC counts ($232 + 205 + 145 + 110 + 60 = 752$) exceeds the 717 excluded papers by 35, corresponding to papers that triggered more than one EC simultaneously. EC3 and EC1 are the two largest reasons in this exclusion list. That is a property of the list. It does not confirm the family codes in Section 5.1.

4.4. Full-Text Extraction and PDF-Based Triangulation

Structured data extraction was performed on the full 169-study set using the extraction template defined in Table 9. For each study, the LLM extracted the research question, study type, research contribution, PM technique, stochastic technique, software artifact, software process, data source, dataset availability, tool used, main finding, limitations, and replication-package status. Title-and-abstract metadata was used as the primary input for studies without locally available PDFs (94/169); for the remaining 75/169 studies a locally accessible PDF was available and used to triangulate against the LLM-extracted fields. Of these 75 PDFs, 73 were fully coded in the operational triangulation spreadsheet used for the figures and positioning tables below.

The 73-paper PDF triangulation subset is therefore not a separate selection stage and does not override the PRISMA counts. It does not confirm the fields behind F1–F5. On 31 studies the exact match between a human rater and the model on the free-text fields is 0%. F1, F3, and F4 were not re-marked in this revision.

Figure 4 summarizes the operational triage profile of the 73 coded PDFs (54 confirm-include, 13 confirm-include

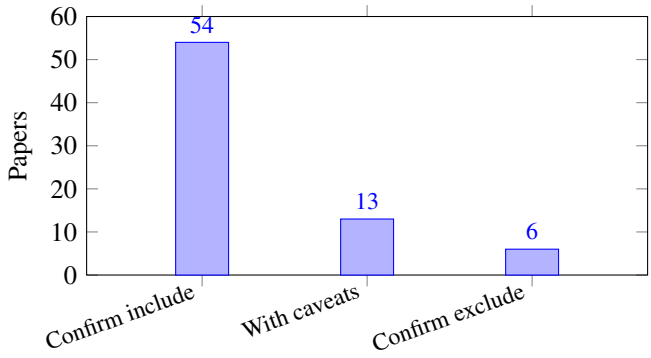


Figure 4: Reading of the 73 PDFs: 54 confirm-include, 13 confirm-include with caveats, and 6 confirm-exclude. These labels are not the screen decisions Include, Maybe, and Exclude. The PRISMA flow is unchanged.

with caveats, 6 confirm-exclude after full-text inspection); the formal selection decisions reported in the PRISMA flow (Figure 1) are unchanged.

4.5. RQ1: Application Landscape

4.5.1. RQ1.1: SDLC Phases Covered

The phase field on the 169 extraction was not recomputed after 17 studies and two extra identifiers left that cut. Those phase percentages are not reported. The family codes that are reported use the 150 studies that remain. They are not a description of the 455.

The phase field was not re-marked.

4.5.2. RQ1.2: Event Log Sources and Construction

The source families, the public-dataset count, and the replication-package count are free-text fields. On 31 studies the exact match between a human rater and the model on those fields is 0%, and the fields were not re-extracted. F2 is withdrawn. Those percentages are not reported, and they do not motivate RA1.

A complementary body of work constructs Markov models directly from GitHub repository traces, bypassing explicit process discovery in favour of manual state-space definition. Jo et al. [27] modelled developer activities in GitHub repositories as Discrete-Time Markov Chains (DTMCs), finding that commits account for 62.73% of long-term activity, and employed probabilistic temporal logic to extract collaboration patterns relevant to project management. This approach was extended in [28], where Probabilistic Model Checking with Computation Tree Logic was applied across five repositories to characterise temporal dynamics, state-transition probabilities, and key behavioural drivers, yielding transparent and explainable project forecasts. Ortu et al. [29] complemented these structural analyses by combining Markov chains with social network analysis to identify fault-insertion and fault-fixing behavioural patterns across Apache Software Foundation projects, demonstrating that developer expertise significantly moderates defect introduction likelihood. Collectively, these studies confirm that SDLC event logs extracted from public repositories are sufficiently rich to support state-transition modelling. However, the event-to-state mapping adopted across

these works is defined manually and differs from study to study; none derives the state space from a discovered process model.

4.6. RQ2: Techniques and Prediction

4.6.1. RQ2.1: Discovery Algorithms Applied to SDLC

The canonical origin of process mining applied to software development contexts can be traced directly to the foundational contributions of Cook and Wolf in the mid-1990s. In their seminal work, [2] introduced three automated process discovery methods—algorithmic grammar inference, Markov models, and neural networks—demonstrating that finite-state machine representations of software processes could be extracted automatically from event traces, substantially lowering the barrier to formal process modeling. This line of inquiry was consolidated in [3], which refined the event-based discovery methodology and established the conceptual vocabulary of inferring workflow structures from historical execution data, a paradigm that remains central to the field today. Notably, both contributions embedded stochastic reasoning—particularly Markov chain representations—within the discovery pipeline, anticipating the probabilistic extensions that would later characterise mature process mining frameworks. The translation of these principles into an explicit, named framework for software engineering contexts was accomplished by [4], who applied the ProM toolkit to source code management logs and demonstrated multi-perspective analysis spanning control-flow, informational, and organisational dimensions. Together, these three works constitute the seminal arc of TF-PM, establishing automated discovery from SDLC event logs as both technically feasible and analytically productive.

The family-code count on the 150 working-set studies that remain, which this revision did not re-mark, has 103 studies (68.7%) matching TF-PM. The same count has 27 papers (18.0%) matching TF-FC. The names of individual algorithms are free-text fields. They were not re-extracted. Those names are not reported.

TF-ST is 49 of these 150 (32.7%). The other 101 do not carry that code, and 49 + 101 = 150. The column `stochastic_technique` is not that code. On these 150 rows it is `none` for 96 and a named technique for 54, and 96 + 54 = 150. Twenty-six of the 54 also contain the token `other`. Six rows coded TF-ST have `none` in the column, and 11 rows with a named technique are not coded TF-ST. On the original 169 the same column was `none` for 110 rows, and 29 rows also contained `other`. Set beside TF-ST = 52, those two figures do not partition 169: 52 + 110 = 162 and 110 + 29 + 52 = 191. The four technique bars in Figure 5 are tokens in that column. They are not a partition of the 49.

4.6.2. RQ2.2: Predictive and Stochastic Techniques

Three distinct sub-families of stochastic techniques emerge from the confirmed literature. The first, **Markov-chain-based prediction**, encompasses models that represent software processes as discrete state transitions, including reinforcement learning agents grounded in Markov Decision Processes for pull-request outcome prediction [30], probabilistic risk quantification from repository data [31], defect removal effectiveness under code churn [32], agile

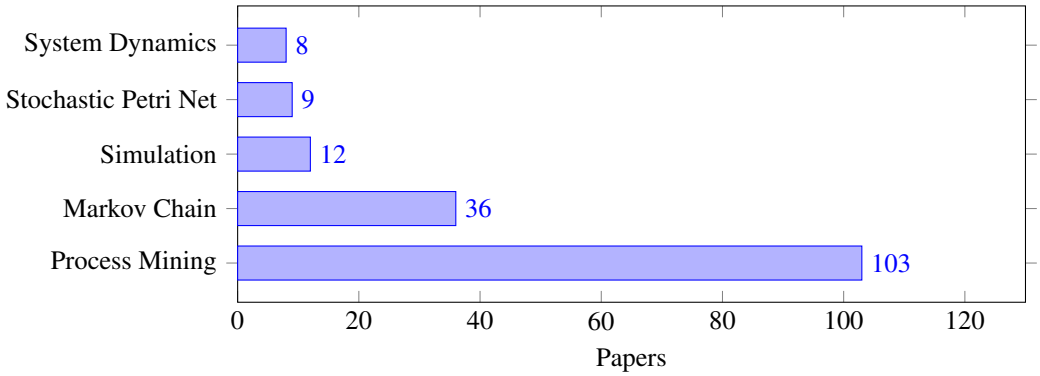


Figure 5: Process mining is the TF-PM family code on the 150 working-set studies that remain. The other four bars are tokens in `stochastic_technique`. They are not a partition of TF-ST, and a paper may carry more than one token. These counts were not re-marked.

state-space planning and tracking [33], release-time forecasting via generalised reliability models [34], and Stochastic Automata Networks for Waterfall team estimation [35]. The second sub-family, **stochastic Petri nets**, models concurrency and asynchrony in DevOps and CI/CD pipelines through non-Markovian specifications [36] and sensitivity analyses that identify bottleneck stages governing delivery probability [37, 38]. The third sub-family, **Monte Carlo simulation**, drives project-level forecasting in hybrid System Dynamics models [39], agile risk assessment frameworks [40], cycle-time economic analysis [41], and the PRIMAD digital-twin architecture for SDLC [42].

Despite their methodological diversity, the cited descriptions share a parametrisation pattern. Transition probabilities, firing rates, and distributional parameters in these descriptions come from aggregate project statistics—historical defect counts, phase-level duration averages, or summary velocity metrics—rather than from transition frequencies extracted through process mining of fine-grained event logs [43, 37, 39]. In the same descriptions, model states stay anchored to coarse lifecycle phases. This is a reading of those descriptions. It is not a re-mark of the family codes labeled F3, which are the counts on the 150 and the 292.

Three studies exemplify the productive convergence of process mining and stochastic modelling at the component level. Incerto [44] demonstrated that variable-length Markov chains outperform eighteen existing stochastic conformance-checking techniques across ten of eleven benchmark datasets, establishing that higher-order sequential dependencies in event logs can be captured without sacrificing computational tractability. López-Pintado [45] extended process-simulation fidelity by replacing deterministic resource-availability calendars with probabilistic counterparts discovered directly from event logs, showing that stochastic resource representations more accurately replicate empirical cycle-time distributions. Jalote [46] took a complementary perspective, modelling programmer task sequences as first-order Markov chains derived from IDE interaction logs and demonstrating that transferring the process patterns of high-productivity developers to average-productivity peers yields measurable productivity gains. In the studies cited in this paragraph, the stochastic layer stays on one analytical concern. That reading does not establish a unified forecasting pipeline in the literature.

The closest published work to a fully integrated multi-stage architecture is PRIMAD, proposed by Guinea-Cabrera [42], which orchestrates pm4py-based process discovery, Akka actor-system simulation, and LightGBM supervised learning within an evolutionary Digital Twin framework for the SDLC. Nevertheless, PRIMAD remains at what may be characterised as Level 2 integration: validation is reported per component rather than across the assembled pipeline, no formal inter-stage contracts govern the handoffs between discovery, simulation, and learning modules, and the architecture incorporates no dedicated Monte Carlo forecasting layer capable of propagating uncertainty estimates end-to-end. On the family codes filled for the 379 studies, PRIMAD is the one row that matches the three families. The 338 were not re-marked in this revision. None of the 74 added studies read from an abstract or a PDF adds a second triple. The 76 title-only studies are outside that count. A single code on that cut is not a statement that the literature contains no end-to-end pipeline.

Studies that combine process mining with forecasting objectives (TF-PM ∩ TF-FC) tend to approach prediction through *process enhancement* rather than through explicit stochastic modelling. Gupta [47] demonstrated that combining inductive miner discovery with Markov chain-based predictive analytics enables identification of recurring failure patterns in software maintenance workflows. Pourbafrani et al. [48] presented SIMPT, a tool that extracts process trees with time parameters from event logs and enables interactive what-if simulation; its successor work [49] further integrates generative AI with business process simulation to produce conformance-preserving what-if scenarios. Caldeira et al. [50] mine IDE interaction logs and show that process complexity metrics derived from ProM correlate moderately ($r \approx 0.43$) with software cyclomatic complexity, achieving predictive accuracies exceeding 92%, demonstrating the feasibility of process-derived features for product-quality prediction.

Taken together, the papers cited in this paragraph describe PM output feeding a downstream model. That reading is not a census of the SE literature. None of these cited systems is described as reporting a probability distribution over completion time or remaining steps for an individual process instance.

4.7. RQ3: Integration and Gaps

4.7.1. RQ3.1: PM and Stochastic Integration Level

The family codes on the 150 working-set studies that remain, which this revision did not re-mark, place rows on the L0–L3 scale defined in Table 15: the number of technique families TF-PM, TF-ST, and TF-FC a study matches. The largest two-family cluster, comprising papers that touch TF-ST and TF-FC without TF-PM—including stochastic Petri net models of CI/CD pipelines [37, 38], Markov-based project estimation [35, 33], and Monte Carlo risk simulation [40, 41]—is coded **L1**, applying stochastic modelling to software processes without any process discovery layer. Papers spanning TF-PM and TF-ST, such as the VLMC conformance checker of [44] and the probabilistic resource-calendar discoverer of [45], are also coded **L1**: they couple discovered process structure with stochastic reasoning, but neither closes the loop toward distributional schedule forecasting. Papers combining TF-PM with TF-FC, such as [47] and [50], use process mining descriptively to assess workflows but lack the stochastic machinery required for Monte Carlo distributional forecasts. The PRIMAD framework [42] is the one triple match in these codes. It assembles

event-log discovery, simulation, and ML prediction, and its evaluation is described as component-wise, so the code is L2 rather than a validated end-to-end contract. No row in these 150 codes is **L3**. L3 would be a row that matches the three families and states formal interfaces validated end-to-end on SDLC event logs. These codes do not establish that no such architecture exists in the literature.

The same 150 codes, not a confirmation of a literature pattern, are as follows. 1 of 150 papers (0.7%) jointly satisfies TF-PM, TF-ST, and TF-FC. Pairwise codes are TF-PM∩TF-ST in 8 papers (5.3%), TF-PM∩TF-FC in 5 papers (3.3%), and TF-ST∩TF-FC in 18 papers (12.0%). No row in these 150 codes is an **L3** architecture. The same triple on the 292 joint-table studies is also one paper. A rare code on these cuts is not a rare pattern in the literature. Figure 6 is the reading priority of the 73 PDFs. It is not the L0–L3 scale of Table 15.

4.7.2. RQ3.2: What the codes do not establish

Two stored readings of the 150 working-set studies that remain follow. First, the family-code count labeled F1: 103 papers (68.7%) match process mining and 27 (18.0%) match forecasting of a process metric. TF-PM with neither TF-ST nor TF-FC is 91 (60.7%). Second, the model assigned study types on these 150 rows: 66 (44.0%) case studies, 29 (19.3%) tool papers, 26 (17.3%) theoretical, 12 (8.0%) simulation-only, 8 (5.3%) controlled experiments, 5 (3.3%) surveys, and 4 (2.7%) residual categories (mixed, framework, exploratory; sum: $66 + 29 + 26 + 12 + 8 + 5 + 4 = 150$). That assignment was not re-marked. F2 is withdrawn: on 31 studies the exact match between a human rater and the model on the free-text fields is 0%, so the public-dataset and replication-package percentages are not reported. The scores in Table 8 are what the same model assigned. They do not stand in for that count.

These readings do not establish that an end-to-end workflow is absent from the literature. The evidence-type distribution in the 73-PDF subset (Figure 7) is a view of that subset. It is not a separate selection layer, and it does not confirm the fields.

5. Discussion

5.1. Coded counts that are not findings

F2 and F5 are withdrawn. On 31 studies a human rater and the model had 0% exact agreement on the free-text fields, and those fields were not re-extracted, so no percentage from them is reported. F1, F3, and F4 are stored family-code counts. Each sentence names its cut: 150, 292, or the 379 for the triple. None of those percentages is a rate over the 455, none was re-marked, and none is a finding. None of the five motivates RA1–RA4.

F1: Stored family codes on 150 and on 292. In the **working-set cut** (150 papers), 103 (68.7%) match TF-PM and 27 (18.0%) match TF-FC. TF-PM with neither TF-ST nor TF-FC is 91 (60.7%). In the **joint cut** (292 papers), 138 (47.3%) match TF-PM, 96 (32.9%) match TF-FC, and 122 (41.8%) match TF-PM with neither TF-ST nor TF-FC. The shift between the two cuts is a difference in which rows the auxiliary corpus added. It is not a claim about the literature.

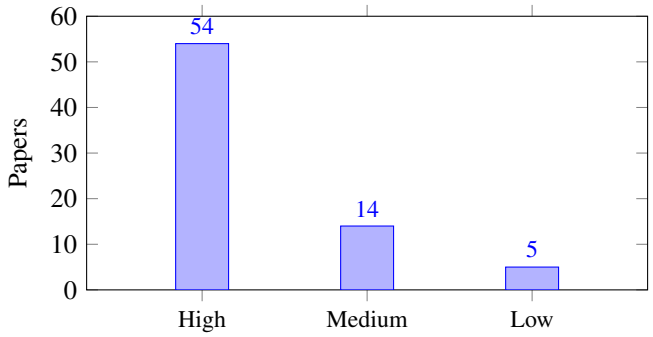


Figure 6: Reading priority of the 73 PDFs. High, Medium, and Low order those PDFs for reading. They are not the integration levels L0–L3 in Table 15 (121, 28, 1, and 0 on the 150 studies).

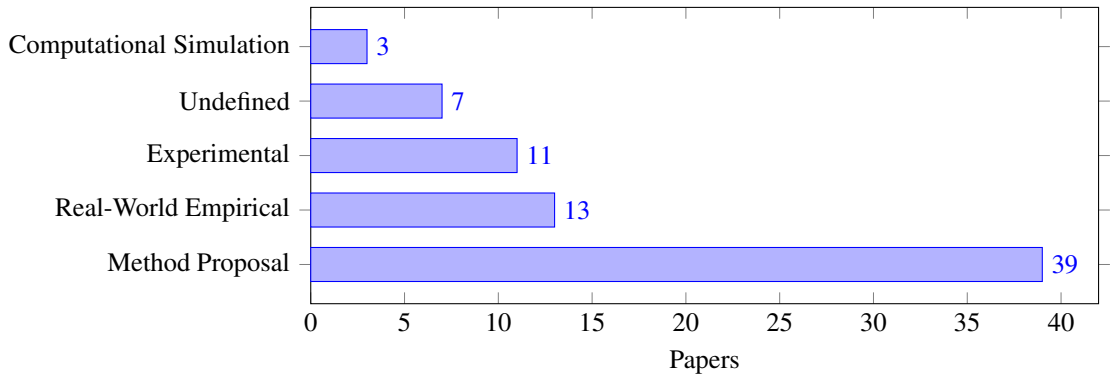


Figure 7: Evidence-type distribution in the 73-paper PDF triangulation subset.

F2: Withdrawn. Event-log construction, dataset availability, and replication-package status are free-text fields. Exact agreement with a human rater is 0% on the 31-study sample, and the fields were not re-extracted. No percentage is reported. F2 does not motivate RA1. The quality scores are assigned by the same model. They do not replace this item.

F3: Stored stochastic family codes on 150 and on 292. Working-set cut: 49 of 150 papers (32.7%) match TF-ST, and the other 101 do not (49 + 101 = 150). TF-ST∩TF-FC covers 18 papers (12.0%). TF-PM∩TF-ST covers 8 (5.3%). Joint cut: 152 of 292 (52.1%) match TF-ST, and the other 140 do not (152 + 140 = 292). TF-ST∩TF-FC covers 80 (27.4%). TF-PM∩TF-ST covers 10 (3.4%). The joint cut contains more rows with TF-ST and without TF-PM than the working-set cut. That is a difference between the two cuts. It is not a description of the stochastic literature. The largest second-pass exclusion in the working-set queue is EC3 (232 of 886). That is a property of this screen. The technique-name column is not added to these family codes. On the original 169 it does not partition the rows together with TF-ST, as stated above.

F4: Stored pairwise and triple codes on 150, 292, and 379. The family marks added in this revision are the 74 studies read from an abstract or a PDF that meet a content criterion, inside the 379. None of those 74 has all three families. The code file does not record a formal PM → Markov → MC → ML contract. That is a property of the file. It is not a finding that no such contract exists. Working-set cut (150 papers): the joint convergence TF-PM∩TF-ST∩TF-FC appears in 1 paper (0.7%). Pairwise convergences are TF-PM∩TF-ST: 8 (5.3%), TF-PM∩TF-FC: 5 (3.3%), TF-ST∩TF-FC: 18 (12.0%). Joint cut (292 papers): TF-PM∩TF-ST∩TF-FC remains 1 paper (0.3%), with pairwise convergences TF-PM∩TF-ST: 10 (3.4%), TF-PM∩TF-FC: 7 (2.4%), TF-ST∩TF-FC: 80 (27.4%). These counts describe these cuts, not the 455-study corpus. Recall on the working-set sample is 0.630 (29 of 46; Wilson 95% CI 0.486–0.755). That figure is not a full-text Recall: a PDF was read for 0.5% of the 886 records. The screen missed eligible studies, so a rare pattern in the 338 is not a rare pattern in the literature. PRIMAD [42] is that one study inside the 379. It assembles discovery, simulation, and learning components but validates each independently and leaves the unified pipeline unevaluated. Its scope is SDLC-oriented: it builds a digital twin from issue-tracking flows in JIRA. It combines empirical process discovery (`pm4py`), a Python-based simulation core, an Akka actor system, and supervised machine learning for context-aware prediction. The reported results are component-wise: the paper does not validate a coupled PM → stochastic simulation → ML pipeline as one architecture. The 76 added studies read from the title alone are outside this count.

F5: Withdrawn. Process-derived features and the coupling of a named mining algorithm with prediction are free-text fields. On 31 studies the exact match with a human rater is 0%, and the fields were not re-extracted. No percentage is reported, including the share of ML rows that name a process-model feature. F5 does not motivate RA3. The 19 IC4c title hits from the Scopus search of 31 October 2026 are reported in Section 2.2. They are not a substitute for this item.

5.2. Integration Levels

Table 15 defines the integration levels used throughout the article and reports their counts. A level is determined only by how many of the three *technique* families a study matches (TF-PM, TF-ST, TF-FC); TF-MSR describes the data source and is not counted. With this single definition, the L1 and L2 rows coincide with the pairwise and triple intersections reported under F3 and F4.

Table 15: Integration levels of the 150 working-set studies that remain. A level counts how many of TF-PM, TF-ST, and TF-FC the study matches.

Level	Operational definition	Papers	Example
L0	At most one technique family	121	Single-technique studies
L1	Exactly two technique families (TF-PM∩TF-ST: 7; TF-PM∩TF-FC: 4; TF-ST∩TF-FC: 17)	28	Incerto [44]; López-Pintado [45]
L2	All three technique families (TF-PM∩TF-ST∩TF-FC)	1	PRIMAD [42]
L3	All three families coupled through explicit interfaces and validated end-to-end	0	—
Total		**150**	

Note: L1 counts exclude the single L2 study, so they are one lower than the inclusive pairwise intersections in F3/F4 (8, 5, 18). The three exclusive pairs sum to 28 (7 + 4 + 17). Figure 6 reports reading priority for 73 PDFs. It is not this L0–L3 scale.

The authors' PATHCAST proposal is one candidate design for L3. It is **not** evaluated in this mapping study and is listed only as a research-agenda hypothesis (RA2–RA4, Section 5.4). It is not inferred from F1–F5.

5.3. Proposed Taxonomy: SPMF

The authors propose the Software Process Mining and Forecasting (SPMF) taxonomy. It is not inferred from F1–F5. The three dimensions below are a proposal by the authors:

Analytical Depth classifies studies along four levels (labelled AD to avoid confusion with the integration levels L0–L3): AD1 (Descriptive: discovery, conformance), AD2 (Diagnostic: bottleneck, rework), AD3 (Predictive: outcome, remaining time), and AD4 (Forecasting: distributional, probabilistic).

SDLC Coverage maps which lifecycle phases are addressed: Requirements and Design, Development (coding, commits, PRs), Quality (testing, reviews, bugs), Delivery (CI/CD, release), and Operations (incidents, maintenance).

Technique Integration classifies methodological combination: Single (PM only, ML only, MC only), Dual (PM + ML, PM + Markov, Markov + MC), and Pipeline (PM → Markov → MC → ML).

5.4. Research Agenda

The four directions below are proposals by the authors.

RA1: Standardized Event Log Construction for SDLC. Not inferred from F2. Target: frameworks for transforming heterogeneous repository data into comparable event logs with cross-tool case correlation.

RA2: Process-Aware Stochastic Forecasting. Not inferred from F3. Target: formal integration of mined transition structure with Markov/Monte Carlo forecasting.

RA3: Process-Derived Features for ML. Not inferred from F5. Target: systematic use of conformance, rework, and path-entropy features in SE predictors.

RA4: Empirical Benchmarking of Integrated Pipelines. Not inferred from F4. The authors' reading of PRIMAD is reported under F4 as a reading of that paper, not as a count of the literature. Target: comparable OSS benchmarks with temporal hold-out for end-to-end PM–stochastic–ML pipelines. Formal specification and evaluation of candidates such as PATHCAST remain future work by the authors.

5.5. Implications for Practitioners

No practitioner recommendation is drawn from these counts. The readings of individual cited papers in Section 4.6 are readings of those papers. They are not a recommendation derived from the 169 codes.

6. Threats to Validity

6.1. Construct Validity

The search strings may exclude studies that use non-canonical terminology. This threat is mitigated through three mechanisms: the complementary MSR string, the frontier stochastic-PM string, and one snowballing iteration over 595 records attributed to 400 source DOIs (Section 2.2). The ten control papers were recovered, but this is a weak check: they were known when the strings were written, so their recovery does not bound the proportion of relevant studies the strings miss. Those ten papers are not the snowballing seed list.

6.2. Internal Validity

LLM-based screening introduces the risk of systematic decision bias. We report six aspects of how this risk was handled: (i) structured outputs; (ii) author inspection of `include` decisions; (iii) human validation of the selection against a two-rater reference standard; (iv) LLM–LLM cross-model agreement; (v) the T/A false-positive rate; and (vi) the closure rule for unresolved cases used in the original run.

(i) **Structured outputs**. All screening responses use a strict JSON schema, enabling post-hoc analysis of decision patterns and rationale distributions.

(ii) **Author inspection of LLM `include` decisions**. Every paper retained at the FT stage of the original run was inspected by the first author against the PRISMA flow before being added to the 169-study working-set tier. The escalation set was judged by both reviewers (Section 6).

(iii) **Human validation of the selection**. *T/A stage.* A stratified random 20% sample of working-set T/A decisions (n = 468) was screened by a human rater blind to the LLM. In the working set, LLM `maybe` decisions were forwarded to FT screening (Section 3); against the human labels, the operational T/A Recall is therefore 72/72 = 1.000 (Wilson 95% CI 0.949–1.000). Counting `maybe` as a negative, as the R1 manuscript did, gives 18/72 = 0.250, but that figure describes a decision rule the working set never applied.

FT stage — R1 reference standard withdrawn. The FT reference standard reported in R1 (n = 177) did not record eligibility: all 126 human `include` labels coincided with papers for which a PDF had been obtained, and 50 of the 51 human `exclude` labels were annotated "PDF not accessible". The resulting Recall (0.246) measured PDF availability, not screening error, and is withdrawn.

Second pass — round-2 validation. The replication files label this pass FT. Before any new judgement we pre-registered the check (`PREREGISTRATION_round2_validation.md`; success if the point estimate of Recall is at least 0.90). The same 177 records were judged against Table 6 by two raters, independently, blind to the LLM and to each other. Disagreements were resolved by consensus. A record whose text could not be obtained is `not_assessable` and leaves the denominator. Six records are in that class. Agreement between the raters, before consensus, is κ = 0.654.

The reference standard is the consensus after a full-text re-reading of the 23 records the first consensus had included and the LLM had excluded. Five of those 23 are ineligible. One is not assessable, because the file on disk is a conference programme. Seventeen remain eligible. Against that reference the LLM's Recall on the working-set sample is 0.630 (29 true positives out of 46 eligible studies; Wilson 95% CI 0.486–0.755). This is not a full-text Recall. A PDF was read for 0.5% of the 886 records, and the second-pass prompt received the title and the abstract. Lost evidence is 0.370. MCC is 0.656 and WMCC with false negatives weighted ten to one is 0.168 (Table 16). The same Recall, 0.630, is obtained if the second rater's labels are used alone. The first author had computed the round-1 agreement report on this sample and may remember some LLM decisions. The second rater had no such exposure. The result is below 0.90, so the pre-registered rule required a human re-screen of the accessible working-set records the LLM had excluded.

Auxiliary tier. The same rule required a check of LLM non-includes outside the working set. Three strata were sampled: S1, title/abstract excludes (N = 2,423, sample 40); S2, full-text excludes (N = 537, sample 40); S3, records closed without a decision under the extended exclusion rule (N = 611, sample 60). The first author judged each sampled record. The second author judged a 30% subset (κ = 0.494). Twenty auxiliary includes were mixed into the sheets so the rater could not infer that every row was an exclusion. The sample estimates of missed studies are 60.6 in S1, 110.2 in S2, and 12.0 in S3, each already rounded to one decimal. Added before that rounding they are 182.7. That sample estimator of auxiliary Recall is 0.563 (235 over 235 plus 182.7), which is what triggered the full re-screen.

Both reviewers then screened the working-set exclusions and the rest of S1, S2, and S3, including the records whose first label was heuristic. They agreed with 92% of those suggested labels. The sheets (4,005 rows) store a decision equal to the suggestion on every row, so the labels they did not accept are not marked there. Rebuilt from the title or abstract the LLM treated as sufficient, that share of the coded includes is the 33 studies in `labels_ta_insufficient_for_include.csv` (33 of 338, 9.8%), and those 33 are removed from the corpus. The counts of missed studies below follow the suggested includes. The suggested includes contain 89 studies the LLM had missed (10 in the sample, 33 in the rest of S1, 32 in the rest of S2, 14 in the rest of S3). Auxiliary Recall on that count is 0.725 (235 of 324). It is still below 0.90. Seventeen of the 33 S1 includes, and all 14 S3 includes, began as heuristic suggestions. The reviewers' 92% agreement covers those suggestions. It does not replace them with a second written code on each row. After de-duplication, 168 records were new to the list. The eligibility pass removes 18 of them, so the studies new to the corpus are the 150 in Section 4. The 89 is the auxiliary slice of the screen. The 150 also includes working-set records the LLM had excluded.

Table 16: Round-2 check of the working-set sample. The human label is the consensus reference. Lost evidence is 1 − Recall. The interval is a Wilson 95% confidence interval. Six records judged `not_assessable` are outside this table. The Recall in this table is not a full-text Recall: a PDF was read for 0.5% of the 886 records, and the prompt received the title and the abstract.

	TP	FP	FN	TN	Recall	95% CI	Lost ev.
Working-set sample	29	5	17	120	0.630	(0.486, 0.755)	0.370

The two tables that follow are the round-1 matrices. The row labelled FT (withdrawn) measured whether a PDF was on disk (Recall 0.246; $\kappa = 0.122$). That row is withdrawn. The 0.630 above is the Recall of the working-set sample. It is not a full-text Recall.

QA and extraction. On the 31 papers included by both the R1 human rater and the LLM, LLM QA matched the human rater on 68–100% per criterion; free-text extraction fields showed 0% exact-match agreement. Those fields were not re-extracted by a human. F2 and F5, which use them, are withdrawn.

Table 17: Round-1 agreement (Cohen's κ) between the primary LLM screener (claude-haiku-4-5) and an independent human rater on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [24]. The T/A row is the title/abstract check. The row "FT (withdrawn)" ($\kappa = 0.122$, $n = 177$) uses the same PDF-on-disk labels as the withdrawn Recall 0.246. It is not agreement on the criteria, and it is not the κ of this revision.

Stage	N	κ_{multi} (interpretation)	κ_{binary} (interpretation)
T/A	468	0.250 (fair)	0.335 (fair)
FT (withdrawn)	177	0.122 (slight)	0.122 (slight)

(iv) **Cross-model inter-rater agreement (Cohen's κ).** A separate stratified random 20% sample of T/A ($n_{\text{theoretical}} = 468$ from the 2,340-paper working set) and FT ($n_{\text{theoretical}} = 177$ from the 886-paper FT queue) decisions was inde-

Table 18: Round-1 human-vs-LLM confusion on a stratified random 20% sample (human rater as gold standard; counts, not percentages). Recall and Lost Evidence (1−Recall) use Wilson 95% confidence intervals. The T/A row labelled "maybe→positive" treats LLM `maybe` decisions as positive referrals, matching LLM4SCREENLIT R6. The row "FT (withdrawn)" is not a screening result: Recall 0.246 records whether a PDF was on disk. The Recall of this revision is 0.630 on the working-set sample (Table 16). That 0.630 is not a full-text Recall.

Stage	TP	FP	FN	TN	Recall	95% CI	Lost Ev.
T/A (strict)	18	4	54	392	0.250	(0.164, 0.361)	0.750
T/A (maybe→pos.)	72	105	0	291	1.000	(0.949, 1.000)	0.000
FT (withdrawn)	31	3	95	48	0.246	(0.179, 0.328)	0.754

pendently re-rated by a second model from the same provider (*claude-sonnet-4-6*). Of the 468 T/A papers in the sample, 420 produced a parseable decision from the verifier and entered the agreement computation; the remaining 48 papers returned a non-parseable response because the abstract was absent, and the verifier then refused a categorical answer. Those 48 are part of the verifier workload: 48 of 468 sampled title/abstract papers. They stay out of κ because the verifier returned no category. On the 420 parseable papers the binary observed agreement is 410 of 420. If each of the 48 is scored as a disagreement, that observed agreement is 410 of 468. They are not added to Lost Evidence. Lost Evidence is the human false-negative rate on the pre-registered sample, and these 48 are verifier responses with no category. The reported κ is the agreement on the 420 papers that produced a parseable decision, out of 468 sampled. All 177 FT sample papers produced a parseable decision. Both the multi-class κ (over the original `include`/`exclude`/`maybe` or `include`/`exclude`/`pending` categories) and the binary κ (collapsing the rater outputs to `include` versus `not-include`) are reported in Table 19. The binary κ is reported as a secondary signal: both binary κ values (0.695 for T/A and 0.694 for FT) exceed the substantial-agreement threshold $\kappa \geq 0.61$ [24], but two LLMs from the same provider may share systematic biases that inflate LLM–LLM agreement relative to an independent human check. The multi-class FT κ is lower (0.479, moderate) because the verifier returned `pending` on 25/177 FT papers where the primary screener committed to a binary decision.

Table 19: Inter-rater agreement (Cohen's κ) between primary screener (claude-haiku-4-5) and verifier (claude-sonnet-4-6) on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [24].

Stage	N	P_o	P_e	κ_{multi} (interpretation)	κ_{binary} (interpretation)
TA	420	82.1%	50.2%	0.641 (substantial)	0.695 (substantial)
FT	177	76.3%	54.5%	0.479 (moderate)	0.694 (substantial)

The cross-model verification protocol was also extended to the auxiliary tier (FT stage, $n = 276$ stratified random 20% sample). On the auxiliary tier the verifier (Sonnet 4.6) returned `pending` on every paper that the primary screener (Haiku 4.5) had rated as `include`, yielding a multi-class $\kappa = 0.004$ and a binary $\kappa = 0.000$ (Po = 84.8%). For the auxiliary-tier T/A stratum ($n = 39$), Po = 97.4% with $\kappa = 0.000$— a base-rate/prevalence paradox, not evidence

of disagreement between raters when one class dominates. The auxiliary records were afterwards screened by both reviewers, who agreed with 92% of the suggested labels (Section 6). The combined-tier counts still rest on the family codes of the original coded set, plus the triple-intersection count on the 379. They were not re-marked in this revision.

Table 20: Auxiliary-tier inter-rater agreement (cross-model verification: Haiku 4.5 primary vs. Sonnet 4.6 verifier on a stratified 20% sample).

Stage	N	P_o^{multi}	P_o^{binary}	κ_{multi}	κ_{binary}
TA	39	97.4%	97.4%	0.000 (slight)	0.000 (slight)
FT	276	54.3%	84.8%	0.004 (slight)	0.000 (slight)

(v) **T/A false-positive rate**. The LLM confirmatory pass at FT level (Round 1) rejected 33 of 108 T/A `include` papers (30.6%), primarily for EC3 (algorithm without SE evaluation). This false-positive rate is consistent with the T/A-screening false-positive rates reported by Khraisha et al. [23] for medical SLRs (in the 25–35% range, depending on domain specificity) and by Syriani et al. [22] for SE SLRs. A false positive at title/abstract can be removed at the second pass. A false negative is not removed by that pass. Recall on the working-set sample is 0.630. That figure is not a full-text Recall: a PDF was read for 0.5% of the 886 records. Missed studies would affect any claim drawn from the coded rows. This revision draws none from F1–F5. The Round 2 screening reprocessed 495 `maybe` papers with enriched abstracts and brought the working-set tier to 169. The corpus of this revision is the 455 in Table 14.

(vi) **Original closure of unresolved uncertainty (revised in resubmission)**. In the original execution run, papers where metadata remained insufficient or verifier disagreement could not be resolved were closed under EC5-extended, biasing toward false negatives. In the working set this rule affected only records that remained abstract-insufficient at FT; in the auxiliary tier it closed 611 records (stratum S3), whose share of relevant studies is estimated in item (iii). The revised protocol routes `maybe`/`pending` to human review instead (Section 2.4).

6.3. External Validity

The review is limited to English-language peer-reviewed publications, which leaves out work in other languages. The broad time span (January 1994 to 12 April 2026) and five-database coverage mitigate geographic and temporal bias. Grey literature is excluded. The stop rule is the document type at search time: journal articles, conference papers, and workshop papers. Theses, preprints, and technical reports were not searched in a separate round. The auxiliary identifier list (3,807 rows; Section 3) was screened in full. It shares 364 papers with the working set. The mapped set is the 455 in Table 14. The working set was built from Scopus, IEEE, and ACM. SpringerLink, Web of Science, and most of the snowballing records were screened in the auxiliary tier, with greater use of abstracts. A difference between a percentage on 169 studies and a percentage on 318 studies can come from that unequal screen.

A four-stage sensitivity check was executed against the auxiliary corpus. First, a random 200-paper subsample was re-screened with the same T/A LLM protocol used for the working set; 12 of 200 (6.0%) returned a T/A `include` decision and 59 (29.5%) returned `maybe`, yielding a point-estimated 228 additional candidate includes if the auxiliary

corpus were screened in full. Given that this estimate exceeded the working-set include count, the second stage was triggered: a screen of all 3,807 auxiliary papers at title/abstract and at the second pass. The result is reported in Table 22: of the 3,807 auxiliary papers, 232 were T/A-included and 1,151 T/A-marked-maybe, of which 212 survived the second pass as `include`. The arithmetic from there to the 455 is the one in Table 14.

In the original run, 16 auxiliary records stayed pending after abstract enrichment, and 595 records still had no abstract. Those records were closed under EC5. In this revision the two reviewers screened the pending and the unclassifiable records (Section 6).

Table 21: Auxiliary-corpus sensitivity check. A random sample of n papers from the deferred auxiliary corpus was re-screened with the same T/A LLM protocol used for the working set.

Quantity	Value
Sample size	200 (seed=20260428)
Auxiliary corpus total	3807
Working-set confirmed includes	169
Sample T/A include rate	6.0% (n_{inc} = 12)
Sample T/A maybe rate	29.5% (n_{maybe} = 59)
Sample T/A exclude rate	64.5% (n_{exc} = 129)
Expected new includes (point estimate)	228
Expected new includes (upper bound, maybe→include)	1351
Sensitivity ratio versus the 169 working-set includes (point)	135.2%
Sensitivity ratio versus the 169 working-set includes (upper)	799.7%

Table 22: Auxiliary-corpus FT-screening outcome (papers that passed T/A on the 3,807-paper deferred corpus).

Outcome	**Count**	**%**
Forwarded from T/A	1383	100.0
Included (new)	212	15.3
Excluded	291	21.0
Pending / unparsed	880	63.6
Combined set, first auxiliary pass (raw, pre-deduplication)	**381**	—

A second external-validity exposure concerns the original 148-paper EC5 stratum (full text inaccessible at first pass). To bound the residual recall risk, an automated open-access (OA) recovery pass was executed against four public discovery services (Unpaywall, Semantic Scholar Graph API, OpenAlex, CORE). The result is summarised

Table 23: Re-FT screening of aux-pending papers that gained an abstract through the post-hoc enrichment cascade.

Outcome	**Count**	**%**
Re-FT include	23	8.1
Re-FT exclude	246	86.3
Re-FT pending	16	5.6
Total re-FT	**285**	100.0

in Table 24: 27 of the 148 EC5 papers (18.2%) returned at least one OA URL during the recovery pass; of those, 17 PDFs were successfully downloaded (11.5% of the stratum). A targeted audit of those 17 files found that 14 OA links resolved to unrelated papers or generic proceedings and therefore remained under EC5, while 3 papers were substantively re-screened and reclassified to EC3. The active PRISMA EC5 count after this audit is thus 145 rather than 148. The remaining 121 papers (81.8%) remain inaccessible and continue to be reported under EC5 in the PRISMA flow. A parallel OA-download pass over the 212 auxiliary includes recovered 49 PDFs (20.5% of 212; the lower yield reflects the publisher-paywall mix of the auxiliary corpus).

Table 24: EC5 recovery attempt against four open-access discovery APIs.

Source	Recovered	% of EC5
Unpaywall	6	4.1%
Semantic_Scholar	6	4.1%
Openalex	6	4.1%
Core	20	13.5%
Any source	**27**	**18.2%**

6.4. Conclusion Validity

F1, F3, and F4 are the original family codes on the cut named in the sentence. This revision did not re-mark those codes and does not use them as findings. The human re-mark in this revision is the triple intersection on the 74 added studies with an abstract or a PDF: none of them joins the three families. F2 and F5 are withdrawn. On 31 studies the exact agreement between the model and a human rater on those free-text fields is 0%, and agreement on the categorical fields is 29–81%. Those categorical fields were not adjudicated. The agenda is not inferred from F1–F5. The two extraction layers are (i) the 169-study working-set extraction (Section 2.6; Table 9), partly checked against the 73-PDF subset, and (ii) the 212-study auxiliary extraction from abstracts. The quality scores are what the same model assigned. A score taken from an abstract is not a quality assessment of the study. The 4/8 line is not a filter

on any statement. Of the 150 working-set studies that remain, 123 were assigned at least 4/8 (mean 4.90 ± 1.36). Of the 212 auxiliary studies, 179 were (mean 5.09 ± 1.43). Of the 292 joint-table studies that remain, 240 were (82.2%, mean 4.96 ± 1.40). Abstract-only scores stay labeled as abstract-only scores.

Table 25: Model-assigned quality scores on the 169 working-set studies and the 212 auxiliary studies (n = 381, before the duplicate removal that yields the 318 joint tables). A score of 4/8 is not a filter.

Tier	N	Mean	SD	≥ 4/8	< 4/8
Working-set	169	4.88	1.40	136	33
Auxiliary	212	5.09	1.43	179	33
Combined	**381**	**5.00**	**1.42**	**315**	**66**

A residual conclusion-validity threat is the LLM-based extraction of PM-technique, stochastic-technique, and SDLC-phase fields. To mitigate this, the 73-PDF triangulation subset was used to spot-check the extracted fields against the corresponding full texts; no systematic disagreement was observed in the spot-check, but human re-extraction on the stratified sample (n = 31 papers included by both human and LLM) shows 0% exact-match agreement on free-text fields and 29–81% on categorical fields (Section 6, item iii).

6.5. Reproducibility and Use of AI

In line with current Elsevier and Springer-Nature editorial policy, this review explicitly declares the use of AI tools in its conduct. The primary screener at title and abstract, including the second pass, is *claude-haiku-4-5-20251001* (Anthropic), invoked through the Anthropic Message Batches API for screening and through the synchronous Messages API for QA scoring. Those calls omit `temperature`; the API default is 1. The cross-model inter-rater verification described above uses *claude-sonnet-4-6* (Anthropic) as the second rater. AI tools are **not** listed as authors of this work and are responsible for none of the editorial judgements; final selection, the decision not to filter on a quality score, taxonomy design, and the wording of the synthesis are the responsibility of the authors.

A complete replication package is deposited at Zenodo (DOI 10.5281/zenodo.21939600). It contains the search strings for all five databases; deduplication and enrichment scripts; screening and QA prompts with JSON schemas; per-stage decision tables; reporting-quality and extraction tables for both corpus tiers; LLM–LLM and human–LLM agreement reports; the 455-study list of this revision (`included_studies_506.csv`, 455 rows); a completed SEGRESS checklist; and a PRISMA flow snapshot. File-level layout is documented in the package README. The deposit is intended to be self-contained: a reviewer can re-run the LLM-assisted protocol on the deposited working set under the same model identifiers.

7. Conclusion

The corpus of this revision is 455 distinct studies (Table 14): 305 that remain after 33 coded studies are removed from the list of 338, plus 150 that remain after the eligibility pass removed 18 of the 168. On the 379 studies with at least one family mark, one study, PRIMAD, matches process mining, a stochastic model, and forecasting together. Seventy-six added studies, read from the title alone, are outside that count.

F2 and F5 are withdrawn: on 31 studies the exact match on the free-text fields is 0%, so those percentages are not reported. The status of F1, F3, and F4 is in Section 5.1. PRIMAD is the one stored triple in the 379. A Scopus search on 31 October 2026 added machine-learning terms and, on titles alone, found 19 further IC4c studies. They are not in the 455, and none joins the three families.

The SPMF taxonomy and the agenda RA1–RA4 (Section 5.4) are proposals by the authors. PATHCAST is future work by the authors.

The primary validity evidence is the pre-registered two-rater check. Operational title/abstract Recall is 1.000 (72 of 72). Recall on the working-set sample is 0.630 (29 of 46; Wilson 95% CI 0.486–0.755). It is not a full-text Recall: a PDF was read for 0.5% of the 886 records. Auxiliary Recall after the stratum re-screen is 0.725 (235 of 324). Both are below 0.90. The replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600).

Appendix A. Included primary studies

This appendix lists the 455 studies. Table A.26 groups them by family. Table A.27 gives the identifier of each study. The file `included_studies_340.csv` still contains the two papers from 1991 and 1992 (`dd71433d`, `37385fbd`) so that removal can be checked.

Table A.27: Identifier list of the 455 studies. The family pattern is the one counted in Table A.26.

Identifier	Year	Role	Family pattern
`87bce30b`	1995	coded set	TF-PM|TF-ST
`6d68591e`	1996	coded set	TF-PM
`50144afd`	1997	coded set	TF-ST|TF-FC
`07e55287`	1998	coded set	TF-PM|TF-ST
`3f8fd941`	1998	coded set	TF-ST|TF-FC
`66d00b19`	1999	coded set	TF-ST|TF-FC
`afd1b7aa`	1999	coded set	TF-ST|TF-FC
`d044ef55`	1999	coded set	TF-ST
`eb05417e`	1999	coded set	TF-ST
`eb526806`	1999	coded set	TF-PM
`f453a92f`	1999	coded set	TF-ST
`46287754`	2000	coded set	TF-ST|TF-FC
`9c255853`	2000	coded set	TF-ST

Table A.27 – continued

Identifier	Year	Role	Family pattern
a4f13cb1	2000	coded set	TF-ST|TF-FC
ab16b112	2000	coded set	TF-ST
f06ebc31	2000	coded set	TF-ST
39e47477	2001	coded set	TF-ST|TF-FC
4155a4da	2001	coded set	TF-ST
50dda3a0	2001	coded set	TF-ST|TF-FC
839d520d	2001	coded set	TF-ST|TF-FC
8ae72547	2001	coded set	TF-ST|TF-FC
b48dde83	2001	coded set	TF-ST
da2f44fa	2001	coded set	TF-ST|TF-FC
f33d3c89	2001	coded set	TF-ST
0b759736	2002	coded set	TF-ST
883e1702	2002	coded set	TF-ST|TF-FC
ad4f84a7	2002	coded set	TF-ST|TF-MSR
b1aabe94	2002	coded set	TF-ST|TF-FC
b75d202e	2002	coded set	TF-ST|TF-FC
39157e2c	2003	coded set	TF-ST
6e4114dc	2003	coded set	TF-ST|TF-FC
78969f65	2003	coded set	TF-ST
3c810db6	2004	coded set	TF-ST
58996946	2004	coded set	TF-ST
64c03c8d	2004	coded set	TF-ST|TF-FC
ab99a53e	2004	coded set	TF-PM|TF-MSR
b37c10c1	2004	coded set	TF-ST|TF-FC
c9512bbb	2004	coded set	TF-ST|TF-FC
2fbf6d36	2005	coded set	TF-ST|TF-FC
3cc5a1cb	2005	coded set	TF-ST
557b4102	2005	coded set	TF-ST
5b51a065	2005	coded set	TF-ST
a20e7c58	2005	coded set	TF-ST|TF-FC
c94fc209	2005	coded set	TF-ST|TF-FC
2c280053	2006	coded set	TF-PM|TF-MSR
416c979c	2006	coded set	TF-ST|TF-FC
48e69187	2006	coded set	TF-PM|TF-MSR
53434e77	2006	coded set	TF-ST|TF-MSR
7766e7b4	2006	coded set	TF-ST
836b5ef0	2006	coded set	TF-ST|TF-FC
8888cfdc	2006	coded set	TF-ST
8e0eec4d	2006	coded set	TF-ST|TF-FC
9f8b9dba	2006	coded set	TF-ST|TF-MSR
ae4f4926	2006	coded set	TF-ST

Table A.27 – continued

Identifier	Year	Role	Family pattern
b5b1402b	2006	coded set	TF-ST|TF-FC
be942093	2006	coded set	TF-PM
c3683c28	2006	coded set	TF-PM|TF-MSR
cdf1f104	2006	coded set	TF-PM|TF-MSR
d50fcf26	2006	coded set	TF-ST
dbcd2435	2006	coded set	TF-ST|TF-FC
df3ad75a	2006	coded set	TF-PM|TF-MSR
ec94ba4d	2006	coded set	TF-ST|TF-FC
3ddbfd94	2007	coded set	TF-PM
747fc068	2007	coded set	TF-ST|TF-FC
9870248b	2007	coded set	TF-ST
ac05d867	2007	coded set	TF-ST
d7b22f54	2007	coded set	TF-ST
e9a40e7a	2007	coded set	TF-ST
0fcd5ece	2008	coded set	TF-ST
2fb4f513	2008	coded set	TF-ST
3c0dd1ad	2008	coded set	TF-ST|TF-FC
40bc718e	2008	coded set	TF-ST
5c43edc7	2008	coded set	TF-ST|TF-FC
7e6b3f9f	2008	coded set	TF-PM
863bae01	2008	coded set	TF-ST|TF-FC
b8b834e2	2008	coded set	TF-ST
deae809b	2008	coded set	TF-ST|TF-MSR
e080b18d	2008	coded set	TF-ST|TF-FC
008caf63	2009	coded set	TF-ST|TF-FC
3b4591cd	2009	coded set	TF-ST|TF-FC
70af8211	2009	coded set	TF-ST|TF-FC
917f238d	2009	coded set	TF-PM
b2c57fd3	2009	coded set	TF-ST|TF-FC
beb83053	2009	coded set	TF-ST|TF-FC
ddeca0e6	2009	coded set	TF-PM
e856dbe4	2009	coded set	TF-PM|TF-MSR
ee5ccaa2	2009	coded set	TF-PM
fc8f325d	2009	coded set	TF-ST|TF-FC
fcc830f6	2009	coded set	TF-ST
0ef6a5b4	2010	coded set	TF-ST|TF-FC
26582491	2010	coded set	TF-ST
3476b5db	2010	coded set	TF-ST|TF-FC
62936834	2010	coded set	TF-ST|TF-FC
6354fc81	2010	coded set	TF-ST|TF-FC
7f497573	2010	coded set	TF-ST|TF-FC

Table A.27 – continued

Identifier	Year	Role	Family pattern
a222ef2c	2010	coded set	TF-PM|TF-MSR
ba2ff831	2010	coded set	TF-ST
bafb58a0	2010	coded set	TF-PM|TF-MSR
c113bf1b	2010	coded set	TF-PM|TF-MSR
c72f823c	2010	coded set	TF-ST
d96a3161	2010	coded set	TF-PM|TF-MSR
266210a7	2011	coded set	TF-ST
2de8a9af	2011	coded set	TF-FC
4316248d	2011	coded set	TF-PM
47ed5bf0	2011	coded set	TF-ST|TF-MSR
51b4ae2b	2011	coded set	TF-ST|TF-FC
57d2445c	2011	coded set	TF-ST|TF-FC
7eaf7eb3	2011	coded set	TF-ST
84315802	2011	coded set	TF-ST
917d9592	2011	coded set	TF-ST
b61ea92e	2011	coded set	TF-ST|TF-FC
c24db118	2011	coded set	TF-PM
c517c93f	2011	coded set	TF-PM|TF-MSR
c7682ec2	2011	coded set	TF-PM|TF-MSR
d098c91d	2011	coded set	TF-PM
0c10d633	2012	coded set	TF-ST|TF-FC
23ec55bd	2012	coded set	TF-ST
326da20f	2012	coded set	TF-ST
66a32124	2012	coded set	TF-ST|TF-FC
a3c061ae	2012	coded set	TF-ST|TF-FC
ae20e68e	2012	coded set	TF-PM
bbc13de4	2012	coded set	TF-PM|TF-MSR
ca00050f	2012	coded set	TF-ST|TF-MSR
d4601a2f	2012	coded set	TF-MSR
f5151d89	2012	coded set	TF-ST|TF-FC
3b614f91	2013	coded set	TF-ST|TF-FC
1363926c	2014	coded set	TF-ST|TF-FC
1de657c1	2014	coded set	TF-PM|TF-MSR
2a7de581	2014	coded set	TF-ST|TF-FC
3231578f	2014	coded set	TF-PM
4cff6b4d	2014	coded set	TF-PM|TF-MSR
542728fe	2014	coded set	TF-ST
6fd30d7e	2014	coded set	TF-PM|TF-MSR
829f8c13	2014	coded set	TF-PM|TF-MSR
932f2a18	2014	coded set	TF-PM
ad250ae5	2014	coded set	TF-PM

Table A.27 – continued

Identifier	Year	Role	Family pattern
c8ea828e	2014	coded set	TF-PM
c9cdb64d	2014	coded set	TF-PM
d1ed6154	2014	coded set	TF-PM|TF-MSR
e58074e0	2014	coded set	TF-PM
e722b1cf	2014	coded set	TF-PM|TF-ST|TF-MSR
f6edde4a	2014	coded set	TF-ST
261f6383	2015	coded set	TF-ST|TF-FC
2647ea07	2015	coded set	TF-PM
2ee3dc10	2015	coded set	TF-PM|TF-MSR
3ac74336	2015	coded set	TF-ST|TF-FC
3ad9cd69	2015	coded set	TF-ST|TF-FC
3d379233	2015	coded set	TF-PM|TF-MSR
508f3b5b	2015	coded set	TF-PM
82b3401d	2015	coded set	TF-PM
927817d5	2015	coded set	TF-ST|TF-FC
aa3ce899	2015	coded set	TF-PM|TF-MSR
b8b5b64e	2015	coded set	TF-ST|TF-FC|TF-MSR
c7ed395a	2015	coded set	TF-ST
d2e1b306	2015	coded set	TF-PM|TF-FC|TF-MSR
e23bbc10	2015	coded set	TF-PM
eea29299	2015	coded set	TF-PM
efbaa4df	2015	coded set	TF-FC|TF-MSR
f356af5c	2015	coded set	TF-ST|TF-FC
fa561e12	2015	coded set	TF-PM|TF-MSR
00ce8cfe	2016	coded set	TF-PM
16e31373	2016	coded set	TF-PM
2b343650	2016	coded set	TF-PM|TF-MSR
37b22a16	2016	coded set	TF-PM
4541f643	2016	coded set	TF-PM
4dc479a6	2016	coded set	TF-PM
4f816525	2016	coded set	TF-PM
632737fb	2016	coded set	TF-ST
7ab40121	2016	coded set	TF-ST|TF-FC
b0f56a95	2016	coded set	TF-ST|TF-FC
b7fd840a	2016	coded set	TF-ST
c42be0aa	2016	coded set	TF-PM|TF-ST
eefc7305	2016	coded set	TF-PM|TF-ST
08d64489	2017	coded set	TF-PM
0d31db29	2017	coded set	TF-PM
1145c10d	2017	coded set	TF-FC
160fface	2017	coded set	TF-PM|TF-MSR

Table A.27 – continued

Identifier	Year	Role	Family pattern
20c17d18	2017	coded set	TF-ST|TF-FC
2e76e921	2017	coded set	TF-PM
3d967450	2017	coded set	TF-ST|TF-FC
3ef399e6	2017	coded set	TF-PM
4c73b15f	2017	coded set	TF-FC
4fd2327e	2017	coded set	TF-ST
b5ea4fdd	2017	coded set	TF-ST
b940a63a	2017	coded set	TF-PM
bc54c3b6	2017	coded set	TF-ST|TF-FC
ca6989c0	2017	coded set	TF-PM
d0e040e1	2017	coded set	TF-PM|TF-MSR
d4a1d75c	2017	coded set	TF-MSR
dcbf27d8	2017	coded set	TF-PM|TF-FC|TF-MSR
ec38f0ba	2017	coded set	TF-PM|TF-MSR
fbc469cf	2017	coded set	TF-FC|TF-MSR
04ab260e	2018	coded set	TF-PM
0a07a816	2018	coded set	TF-PM
223a62e3	2018	coded set	TF-PM|TF-MSR
4082a5b1	2018	coded set	TF-PM
4409efef	2018	coded set	TF-FC
4cb686b6	2018	coded set	TF-ST|TF-FC
4e01ae35	2018	coded set	TF-PM
50d875ff	2018	coded set	TF-PM|TF-FC
5c4c0e7f	2018	coded set	TF-PM|TF-FC|TF-MSR
608b005d	2018	coded set	TF-ST|TF-FC
6f9a5a08	2018	coded set	TF-PM
9e341b7f	2018	coded set	TF-ST|TF-FC
d0c47271	2018	coded set	TF-PM
e12fff7c	2018	coded set	TF-PM|TF-MSR
2d369fe9	2019	coded set	TF-PM|TF-ST
537bbfd0	2019	coded set	TF-ST|TF-FC
542c0019	2019	coded set	TF-PM
588a19f5	2019	coded set	TF-PM
6464a4f1	2019	coded set	TF-PM|TF-MSR
6cb59310	2019	coded set	TF-PM
73df5473	2019	coded set	TF-ST
8ea46c1d	2019	coded set	TF-PM
94e88f90	2019	coded set	TF-ST|TF-FC
9662854d	2019	coded set	TF-ST|TF-MSR
b788b176	2019	coded set	TF-PM|TF-MSR
be899615	2019	coded set	TF-PM

Table A.27 – continued

Identifier	Year	Role	Family pattern
c0352476	2019	coded set	TF-PM
c0a779ae	2019	coded set	TF-PM|TF-MSR
d5aff523	2019	coded set	TF-PM|TF-MSR
e627756c	2019	coded set	TF-FC|TF-MSR
e6f42f43	2019	coded set	TF-PM
f16294f2	2019	coded set	TF-PM|TF-MSR
15c9808b	2020	coded set	TF-PM
340c0c24	2020	coded set	TF-ST|TF-FC
5029f905	2020	coded set	TF-PM|TF-MSR
6e572cbb	2020	coded set	TF-PM|TF-MSR
7cf00385	2020	coded set	TF-PM|TF-MSR
8a97d0cf	2020	coded set	TF-ST|TF-FC
8de6dcfa	2020	coded set	TF-ST|TF-FC
99736057	2020	coded set	TF-PM|TF-MSR
b7022ac2	2020	coded set	TF-PM|TF-MSR
16644fb0	2021	coded set	TF-PM|TF-MSR
2d2d29ea	2021	coded set	TF-FC|TF-MSR
2f713568	2021	coded set	TF-ST|TF-MSR
38694b4e	2021	coded set	TF-ST|TF-FC
483d7725	2021	coded set	TF-ST|TF-FC
61e62f9d	2021	coded set	TF-PM
6f295ec6	2021	coded set	TF-PM|TF-MSR
77fa80cd	2021	coded set	TF-PM
7b0b22ea	2021	coded set	TF-PM|TF-ST
9c4cc898	2021	coded set	TF-PM|TF-ST
9e450f55	2021	coded set	TF-PM|TF-ST
a86b1042	2021	coded set	TF-PM
abfd6b2e	2021	coded set	TF-PM
b123faf2	2021	coded set	TF-FC
c894f613	2021	coded set	TF-PM
db43c22c	2021	coded set	TF-ST
e89188a6	2021	coded set	TF-ST|TF-FC
f36a08fe	2021	coded set	TF-ST|TF-FC
f59b10d5	2021	coded set	TF-FC
ffa8885e	2021	coded set	TF-PM|TF-MSR
0587b264	2022	coded set	TF-PM|TF-MSR
4b34f7a5	2022	coded set	TF-PM|TF-FC
68dbd881	2022	coded set	TF-ST|TF-FC
7eebcd00	2022	coded set	TF-PM
88087b27	2022	coded set	TF-PM
ae9983fa	2022	coded set	TF-ST|TF-FC

Table A.27 – continued

Identifier	Year	Role	Family pattern
b5ffb64d	2022	coded set	TF-PM
b8fc130d	2022	coded set	TF-PM|TF-MSR
e4f9257c	2022	coded set	TF-PM|TF-FC
e941deda	2022	coded set	TF-PM|TF-MSR
eeee1fa8	2022	coded set	TF-PM
f52cb3e1	2022	coded set	TF-PM|TF-MSR
fbe93e3d	2022	coded set	TF-PM
fde0a3b4	2022	coded set	TF-PM
ff916fb9	2022	coded set	TF-FC|TF-MSR
12484962	2023	coded set	TF-PM
140e67e6	2023	coded set	TF-ST|TF-FC
1b7688b7	2023	coded set	TF-PM
21851bba	2023	coded set	TF-ST|TF-MSR
472e7acc	2023	coded set	TF-PM
548627c6	2023	coded set	TF-ST|TF-MSR
56342950	2023	coded set	TF-ST|TF-MSR
5b7858a6	2023	coded set	TF-PM
61cb2257	2023	coded set	TF-ST
667f1705	2023	coded set	TF-ST|TF-MSR
82ce5370	2023	coded set	TF-PM|TF-MSR
a88acaff	2023	coded set	TF-PM|TF-MSR
b68e0602	2023	coded set	TF-PM|TF-MSR
bbfaa5a6	2023	coded set	TF-ST
d65695c6	2023	coded set	TF-ST|TF-FC
df1bee14	2023	coded set	TF-ST|TF-FC
e45c53bb	2023	coded set	TF-ST|TF-FC
f512f749	2023	coded set	TF-ST|TF-MSR
2349bd7b	2024	coded set	TF-PM|TF-MSR
3430f3d0	2024	coded set	TF-ST
373ad20e	2024	coded set	TF-PM
63bfddbd	2024	coded set	TF-PM|TF-MSR
6ff2b7f2	2024	coded set	TF-ST|TF-MSR
7f6a881a	2024	coded set	TF-PM
98edc5c2	2024	coded set	TF-ST|TF-MSR
9ffa3123	2024	coded set	TF-ST|TF-FC|TF-MSR
c7d555aa	2024	coded set	TF-PM
69029db6	2025	coded set	TF-PM
7434525f	2025	coded set	TF-PM|TF-MSR
bf72ff27	2025	coded set	TF-PM|TF-MSR
e3d6d00e	2025	coded set	TF-PM|TF-MSR
ed0ab009	2025	coded set	TF-PM|TF-ST|TF-FC

Table A.27 – continued

Identifier	Year	Role	Family pattern
f2c6bd74	2025	coded set	TF-PM|TF-MSR
03fb6812	2026	coded set	TF-PM|TF-MSR
2c1f1193	2026	coded set	TF-PM|TF-MSR
9ea592cd	2026	coded set	TF-PM
f7a54d4c	2026	coded set	TF-PM
c7a601d8	1994	added	title only, marks blank
6115d05e	1997	added	TF-ST
8dea6481	1997	added	TF-ST
12f7f992	1998	added	TF-ST|TF-FC
85002ffb	1998	added	TF-ST
20a00ebf	1999	added	title only, marks blank
e82e259b	2000	added	TF-ST
13983127	2001	added	title only, marks blank
72329237	2001	added	TF-ST
e24af4e7	2001	added	TF-ST
7f8d2b91	2002	added	TF-PM
dcf53abb	2002	added	TF-ST
e3a9143c	2002	added	TF-ST
e620e55e	2002	added	TF-ST
1c928aae	2003	added	TF-PM
3487bbf5	2003	added	TF-ST
368dc22f	2003	added	TF-ST
4c4a7770	2003	added	title only, marks blank
6b16f9de	2003	added	TF-ST
fd7ca261	2004	added	TF-ST
0232276d	2005	added	TF-ST
190c11c4	2005	added	TF-ST
33361b00	2005	added	TF-ST
6d83b874	2005	added	TF-ST|TF-FC
a31e0556	2005	added	TF-ST
fe1cd2ea	2005	added	title only, marks blank
063967eb	2006	added	TF-ST|TF-FC
117db759	2006	added	TF-ST
24059629	2006	added	TF-ST
66c95d97	2006	added	TF-ST
6795e33c	2006	added	title only, marks blank
b78523e4	2006	added	TF-PM
fe75ced8	2006	added	title only, marks blank
16a201ed	2007	added	title only, marks blank
171085b9	2007	added	title only, marks blank
8e41b716	2007	added	TF-PM

Table A.27 – continued

Identifier	Year	Role	Family pattern
9801135d	2007	added	TF-ST|TF-FC
a401497c	2007	added	TF-ST
aa0da064	2007	added	TF-ST|TF-FC
e07c6819	2007	added	TF-MSR
3ae2f9fd	2008	added	title only, marks blank
6f5e6f89	2008	added	title only, marks blank
ba328b1c	2008	added	title only, marks blank
cd2ca776	2008	added	title only, marks blank
dfb2398b	2008	added	TF-ST
e5fa7505	2008	added	title only, marks blank
e6075fca	2008	added	TF-ST
fb4a2cb3	2008	added	title only, marks blank
fbf7e330	2008	added	TF-ST
0906c3aa	2009	added	title only, marks blank
1d19dd67	2009	added	title only, marks blank
67b85117	2009	added	title only, marks blank
8f1fb42b	2009	added	title only, marks blank
989891c5	2009	added	title only, marks blank
a98f6c54	2009	added	TF-ST
ecd93074	2009	added	title only, marks blank
eeed2f5d	2009	added	title only, marks blank
1d754c2e	2010	added	TF-ST
5f66b602	2010	added	title only, marks blank
86f0c466	2010	added	TF-ST
a210fdd3	2010	added	TF-ST
a9bafd0d	2010	added	title only, marks blank
ca22bfe7	2010	added	TF-ST
cd860b9f	2010	added	TF-ST
04e0e983	2011	added	title only, marks blank
4d088b6b	2011	added	title only, marks blank
5c3f99ea	2011	added	TF-ST
704a0d34	2011	added	TF-ST
7a18f4ee	2011	added	title only, marks blank
9ae84d72	2011	added	title only, marks blank
c4269b60	2011	added	title only, marks blank
c815bcc6	2011	added	TF-ST
e969d9bd	2011	added	title only, marks blank
0b62bf5c	2012	added	title only, marks blank
0e15e613	2012	added	TF-MSR
0e708ae8	2012	added	title only, marks blank
1303e41e	2012	added	title only, marks blank

Table A.27 – continued

Identifier	Year	Role	Family pattern
3f09473c	2012	added	TF-ST
8e68ae72	2012	added	title only, marks blank
da28afcb	2012	added	title only, marks blank
f27117e2	2012	added	TF-ST
050fa5cf	2013	added	TF-ST
60e4fba6	2013	added	title only, marks blank
02cc4c2d	2014	added	title only, marks blank
2f1e25da	2014	added	TF-ST
5fe2fe46	2014	added	title only, marks blank
88ee1d90	2014	added	title only, marks blank
8c42d0fe	2014	added	title only, marks blank
e4c752f3	2014	added	title only, marks blank
e69d723f	2014	added	title only, marks blank
e8daaada	2014	added	title only, marks blank
ed17c20e	2014	added	title only, marks blank
3691dee1	2015	added	title only, marks blank
4eeb076b	2015	added	title only, marks blank
921e8fc3	2015	added	TF-ST
f2c78b19	2015	added	title only, marks blank
f822c54d	2015	added	TF-ST
b269731f	2016	added	title only, marks blank
bce026ce	2016	added	TF-PM|TF-MSR
d85e3fdf	2016	added	title only, marks blank
ed81f924	2016	added	TF-ST|TF-FC
01f8cab1	2017	added	title only, marks blank
6217173e	2017	added	TF-MSR
6f007eeb	2017	added	title only, marks blank
7607982d	2017	added	title only, marks blank
a21c9dff	2017	added	title only, marks blank
e3f1f421	2017	added	title only, marks blank
0d4ebe86	2018	added	TF-PM|TF-MSR
56fcb7c7	2018	added	title only, marks blank
65e5d58c	2018	added	TF-ST
9a40c123	2018	added	TF-FC
6e178383	2019	added	TF-FC
f48e9319	2019	added	TF-ST
d8591f96	2020	added	TF-PM
0999fe57	2021	added	TF-ST
0bb6f873	2021	added	title only, marks blank
165593a8	2021	added	TF-ST
4b58cb90	2021	added	title only, marks blank

Table A.27 – continued

Identifier	Year	Role	Family pattern
8d2c0b03	2021	added	title only, marks blank
cc618a37	2021	added	TF-PM|TF-MSR
04657923	2022	added	TF-PM
12343eec	2022	added	title only, marks blank
1fd95ebc	2022	added	TF-PM
22412f71	2022	added	TF-PM|TF-MSR
61048559	2022	added	title only, marks blank
780addad	2022	added	TF-MSR
7a92806a	2022	added	TF-ST|TF-MSR
83421d8c	2022	added	title only, marks blank
92fa0290	2022	added	title only, marks blank
140c9afd	2023	added	TF-FC
1efe6b8c	2023	added	title only, marks blank
44d6a32e	2023	added	title only, marks blank
76a1870d	2023	added	TF-ST
8e09bd55	2023	added	title only, marks blank
aaef3235	2023	added	title only, marks blank
dea836a1	2023	added	title only, marks blank
e3f89404	2023	added	title only, marks blank
063986d6	2024	added	TF-PM
08a023e2	2024	added	title only, marks blank
10bdcff6	2024	added	TF-FC
2c9ac8c4	2024	added	title only, marks blank
6ef49a7b	2024	added	title only, marks blank
6ffcab24	2024	added	title only, marks blank
7221e872	2024	added	title only, marks blank
d36b0adf	2024	added	TF-PM
dd3b63a3	2024	added	TF-MSR
ecc17ad4	2025	added	title only, marks blank
1a59bb38	2026	added	title only, marks blank
4ba3e996	2026	added	TF-MSR
8851bb5e	2026	added	title only, marks blank

Publications dated 2026 that were indexed by 12 April 2026 remain inside the eligibility window.

Appendix B. SEGRESS reporting checklist

Table B.28 maps each SEGRESS item to the location in this manuscript. Items marked N/A are those SEGRESS declares not required for mapping studies (no synthesis of primary-study outcomes).

Table A.26: Classification of the 455 studies. TF-PM is process mining (IC4a), TF-ST a stochastic model (IC4b), TF-FC forecasting (IC4c), and TF-MSR repository mining (IC4d). The 305 remain from the coded list of 338 after 33 studies were removed because the title or abstract does not support inclusion. The 74 are the added studies read from an abstract or a PDF that meet at least one content criterion. The 76 title-only studies have no family mark.

Family pattern	**305**	**74**	**76**
TF-ST	51	44	–
TF-ST|TF-FC	81	6	–
TF-PM	70	9	–
title only, marks blank	–	–	76
TF-PM|TF-MSR	57	4	–
TF-ST|TF-MSR	15	1	–
TF-FC	6	4	–
TF-MSR	2	6	–
TF-PM|TF-ST	8	–	–
TF-FC|TF-MSR	5	–	–
TF-PM|TF-FC	3	–	–
TF-PM|TF-FC|TF-MSR	3	–	–
TF-ST|TF-FC|TF-MSR	2	–	–
TF-PM|TF-ST|TF-FC	1	–	–
TF-PM|TF-ST|TF-MSR	1	–	–
Total	**305**	**74**	**76**

Table B.28: SEGRESS reporting checklist for this systematic mapping study. Locations refer to the revised manuscript. Status: R = reported; P = partial; N/A = not required for mapping studies under SEGRESS.

Item	Requirement (mapping)	Where addressed	St.
1	Title identifies the report as a systematic mapping study and states the topic.	Title page: "A Systematic Mapping Study with an LLM-Assisted Protocol".	R
2	Structured abstract: background, objective, methods, results, limitations, conclusion.	Abstract (incl. a Limitations clause and human-validation statement).	R
3	Rationale: why the review is needed; gap vs prior secondary studies.	Introduction: states that no prior mapping covers the joint PM × stochastic × forecasting intersection.	R
4	Objectives / research questions.	Table 1; Section 2.1.	R
5	Eligibility criteria, date/language restrictions, handling of on-topic prior reviews.	Table 6 (IC1–IC3 publication filters; IC4a–IC4d content; EC1–EC5). Window 1 January 1994 to the search date, 12 April 2026. A 2026 publication counts only if it was indexed by that date. EC4 excludes off-topic secondary studies.	R
6	Information sources with search execution dates.	Tables 3 and 4; each export is dated 12 April 2026. The three Scopus strings named in Table 11 are printed in Section 2.2.	R
7	Full search strategy (strings, snowballing, completeness checks).	Section 2.2. One snowballing iteration stores 595 records attributed to 400 source DOIs. The ten control papers (Table 5) are not those source DOIs. Recovering the ten known papers does not measure search recall.	R

Continued on next page

Table B.28 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
8	Selection process: phases, number of assessors, disagreement handling, automation, agreement.	Sections 2.4, 6: one LLM screener; two human raters on the pre-registered sample (n = 468 at title/abstract, n = 177 at the second pass). Recall on the working-set sample is 0.630 (29 of 46). It is not a full-text Recall: a PDF was read for 0.5% of the 886 records. The round-1 FT row (Recall 0.246; κ = 0.122) is withdrawn. The two reviewers agreed with 92% of the suggested labels on the escalation set. The re-screen sheets (4,005 rows) match the suggestion on every row. The coded includes whose title or abstract does not support inclusion are the 33 in `labels_ta_insufficient_for_include.csv` (33 of 338, 9.8%), and they are removed.	R
9	Data-collection process: reviewers, independence, automation.	The same model extracts the fields. On 31 studies the exact match with a human rater on the free-text fields is 0%. Those fields were not replaced by a human re-extraction. There is no second human extractor for the full set.	R
10a	Outcomes for which data were sought.	N/A: mapping study does not synthesise primary-study outcomes.	N/A
10b	Non-outcome variables and classification schemes.	Extraction schema: PM technique, stochastic technique, SDLC phase, dataset source, IC combination, reporting-quality scores (QA1–QA8).	R
11	Risk-of-bias methods (optional for mapping studies).	Reporting-quality scores (Dybå & Dingsøyr), not risk of bias. The same model assigns them. A score of 4/8 is not a filter on F1–F5, and no statement uses that cut.	R
12	Effect measures.	N/A (no outcome synthesis).	N/A

Continued on next page

Table B.28 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
13 / 13a	Methods to analyse study characteristics; which studies enter each analysis.	Each sentence names its cut. Family-code counts use 150 working-set studies and 292 joint-table studies. The 169 and the 318 are the earlier cuts, before the removals named in Table 14. The corpus of this revision is 455 (Table 14). Quality scores do not decide which studies enter a count.	R
13b	Data preparation for outcome synthesis.	N/A (mapping).	N/A
13c	Methods for tables, graphs, maps of study characteristics.	Frequency tables, year plot, IC-combination counts, SPMF taxonomy, integration-level Table 15.	R
13d	Methods used to synthesise primary-study outcomes.	N/A (mapping).	N/A
13e	Sensitivity of the auxiliary corpus (not outcome synthesis).	Section 6, Table 21.	P
13f	Heterogeneity of primary-study outcomes.	N/A (mapping).	N/A
14	Reporting-bias / publication-bias methods.	N/A for mapping studies under SEGRESS.	N/A
15	Certainty (GRADE) methods.	N/A for mapping studies.	N/A
16a	Selection results, ideally a PRISMA flow; agreement statistics.	Figure 1 ends at the corpus of 455: 340 distinct LLM confirmations, minus two pre-1994 papers, minus 33 coded studies whose title or abstract does not support inclusion, plus 150 after the eligibility pass (168 remained after deduplication; 18 were then removed). Recall on the working-set sample is 0.630, and it is not a full-text Recall; auxiliary Recall is 0.725. The round-1 FT row (Recall 0.246; $\kappa = 0.122$) is withdrawn.	R

Continued on next page

Table B.28 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
16b	Near-misses (optional for mapping).	EC3 is the modal FT exclusion; a full near-miss citation list is not included (optional).	P
17	Cite each included study and present characteristics.	The included-study list has 455 rows in `included_studies_506.csv` (the filename keeps the earlier count). The 340-row file is kept so the removal of the two pre-1994 papers can be checked.	R
18	Risk-of-bias results per study (optional for mapping).	Quality scores are descriptive, not a risk-of-bias judgement. The working-set scores are in Section 2.5. A score of 4/8 does not add or remove a study.	R
19	Results of individual studies (outcomes).	N/A (mapping).	N/A
20a	Analyses of study characteristics addressing each RQ.	Sections 4–5.1 (RQ1–RQ3, F1–F5).	R
20b–20d	Statistical outcome syntheses, sensitivity, heterogeneity.	N/A (mapping).	N/A
21	Reporting-bias results.	N/A (mapping).	N/A
22	Certainty of evidence.	N/A (mapping).	N/A
23a	Interpretation in the context of other evidence.	Discussion / research agenda (Section 5.4).	R
23b	Limitations of the included evidence.	Optional for mapping; extraction-basis and Lost Evidence discussed in Section 6.	R

Continued on next page

Table B.28 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
23c	Limitations of the review process.	Section 6. The mapping protocol was not pre-registered. The human check was. Two raters; Recall on the working-set sample is 0.630, below 0.90, and it is not a full-text Recall. The round-1 FT row (Recall 0.246; $\kappa = 0.122$) is withdrawn. F2 and F5 are withdrawn (free-text exact match 0% on 31 studies). F1, F3, and F4 were not re-marked and do not motivate RA1–RA4. The re-screen sheets (4,005 rows) match the suggestion on every row. The 33 coded includes whose title or abstract does not support inclusion (33 of 338, 9.8%) are removed.	R
23d	Implications for practice, policy, future research.	Sections 5.5, 5.4 (RA1–RA4). No practitioner claim is drawn from F1–F5. The agenda is proposed by the authors and is not inferred from those codes.	R
24a	Registration (or statement that the review was not registered).	The mapping protocol was not pre-registered. The human check of the screen was, before the round-2 judgements.	R
24b	Protocol availability (optional for mapping).	No standalone protocol deposit; methods are specified in Section 2 and the replication package.	P
24c	Protocol amendments (optional for mapping).	Revised `maybe`/`pending` handling documented as a post-submission protocol change (Section 2.4).	R
25	Sources of financial / non-financial support.	Funding statement (CAPES Finance Code 001).	R
26	Competing interests.	Declaration of competing interest.	R

Continued on next page

Table B.28 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
27	Public availability of data, code, and other materials.	Zenodo DOI 10.5281/zenodo.21939600. The included-study list has 455 rows in `included_studies_506.csv`. The 340-row file is kept for the pre-1994 removal check.	P

Source. Item definitions follow Kitchenham, Madeyski and Budgen, SEGRESS, *IEEE Trans. Softw. Eng.*, 49(3):1273–1298, 2023, doi:10.1109/TSE.2022.3174092.

CRediT authorship contribution statement

Rodrigo Almeida de Oliveira: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Juliano de Paulo Ribeiro:** Methodology, Validation, Investigation, Writing – review & editing. **Edson Emilio Scalabrin:** Conceptualization, Methodology, Supervision, Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Funding

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior — Brasil (CAPES) — Finance Code 001, through a doctoral scholarship granted to the corresponding author within the Graduate Program in Informatics at the Pontifícia Universidade Católica do Paraná (PPGIa/PUC-PR).

Declaration of generative AI in the writing process

During the conduct of this systematic mapping study, the authors used *claude-haiku-4-5-20251001* (Anthropic) as the primary LLM screener at the title/abstract stage and at the second pass, which used the title and the abstract, and as the LLM scorer for the quality-assessment rubric, and *claude-sonnet-4-6* (Anthropic) as the cross-model verifier for inter-rater agreement. All decisions were inspected by the authors; the LLM tools are not listed as authors and bear no responsibility for editorial judgements. Full prompts, JSON schemas, and per-paper decisions are deposited in the replication package (Zenodo DOI 10.5281/zenodo.21939600). The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

A full replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600). The package contains: search strings for all five databases, deduplication and enrichment scripts, screening prompts and JSON schemas, the QA prompt, the cross-model κ verification scripts, all decision CSVs at every pipeline stage, the QA-scored 169-study and 212-study tables, the structured extraction tables (working-set and auxiliary), the inter-rater agreement reports (LLM–LLM and human–LLM), the 455-study list of this revision (`included_studies_506.csv`), the earlier 340-study file kept for traceability, a completed SEGRESS checklist, and a PRISMA flow snapshot.

References

[1] B. A. Kitchenham, L. Madeyski, D. Budgen, SEGRESS: Software engineering guidelines for REporting secondary studies, IEEE Transactions on Software Engineering 49 (3) (2023) 1273–1298. doi:10.1109/TSE.2022.3174092.
URL https://doi.org/10.1109/TSE.2022.3174092

[2] J. E. Cook, A. L. Wolf, Automating process discovery through event-data analysis, in: Proceedings of the 17th International Conference on Software Engineering (ICSE 1995), ACM, 1995, pp. 73–82. doi:10.1145/225014.225021.

[3] J. E. Cook, A. L. Wolf, Discovering models of software processes from event-based data, ACM Transactions on Software Engineering and Methodology (TOSEM) 7 (3) (1998) 215–249. doi:10.1145/287000.287001.

[4] V. Rubin, C. W. Günther, W. M. P. van der Aalst, E. Kindler, B. F. van Dongen, W. Schäfer, Process mining framework for software processes, in: International Conference on Software Process (ICSP 2007), Vol. 4470 of LNCS, Springer, 2007, pp. 169–181. doi:10.1007/978-3-540-72426-1_15.

[5] J. A. Whittaker, M. G. Thomason, A Markov chain model for statistical software testing, IEEE Transactions on Software Engineering 20 (10) (1994) 812–824. doi:10.1109/32.328991.

[6] N. B. Ali, K. Petersen, C. Wohlin, A systematic literature review on the industrial use of software process simulation, Journal of Systems and Software 97 (2014) 65–85. doi:10.1016/j.jss.2014.06.059.

[7] H. Kagdi, M. L. Collard, J. I. Maletic, A survey and taxonomy of approaches for mining software repositories in the context of software evolution, Journal of Software Maintenance and Evolution: Research and Practice 19 (2) (2007) 77–131. doi:10.1002/smr.344.

[8] T. Hall, S. Beecham, D. Bowes, D. Gray, S. Counsell, A systematic literature review on fault prediction performance in software engineering, IEEE Transactions on Software Engineering 38 (6) (2012) 1276–1304. doi:10.1109/TSE.2011.103.

[9] A. E. Márquez-Chamorro, M. Resinas, A. Ruiz-Cortés, Predictive monitoring of business processes: A survey, IEEE Transactions on Services Computing 11 (6) (2018) 962–977. doi:10.1109/TSC.2017.2772256.

[10] I. Teinemaa, M. Dumas, M. La Rosa, F. M. Maggi, Outcome-oriented predictive process monitoring: Review and benchmark, ACM Transactions on Knowledge Discovery from Data 13 (2) (2019) 1–57. doi:10.1145/3301300.

[11] I. Verenich, M. Dumas, M. La Rosa, F. M. Maggi, I. Teinemaa, Survey and cross-benchmark comparison of remaining time prediction methods in business process monitoring, ACM Transactions on Intelligent Systems and Technology 10 (4) (2019) 1–34. doi:10.1145/3331449.

[12] C. d. S. Garcia, A. Meincheim, E. R. Faria Junior, M. R. Dallagassa, D. M. V. Sato, D. R. Carvalho, E. A. P. Santos, E. E. Scalabrin, Process mining techniques and applications – a systematic mapping study, Expert Systems with Applications 133 (2019) 260–295. doi:10.1016/j.eswa.2019.05.003.

[13] M. Thiede, D. Fuerstenau, A. P. Bezerra Barquet, How is process mining technology used by organizations? a systematic literature review of empirical studies, Business Process Management Journal 24 (4) (2018) 900–922. doi:10.1108/BPMJ-06-2017-0148.

[14] H. Zhang, B. Kitchenham, D. Pfahl, Software process simulation modeling: An extended systematic review, in: New Modeling Concepts for Today's Software Processes (ICSP 2010), Lecture Notes in Computer Science, Springer, 2010, pp. 309–320. doi:10.1007/978-3-642-14347-2_27.

[15] P. Mukala, A. Cerone, F. Turini, Process mining event logs from FLOSS data: State of the art and perspectives, in: Software Engineering and Formal Methods (SEFM 2014 Collocated Workshops), Lecture Notes in Computer Science, Springer, 2015, pp. 182–198. doi:10.1007/978-3-319-15201-1_12.

[16] R. Thamizharasan, K. Appavoo, A review on software process mining using Petri nets, Asian Journal of Applied Sciences 9 (3) (2016) 131–142. doi:10.3923/ajaps.2016.131.142.

[17] E. Stepanov, A. Mitsyuk, Continuous software monitoring backed by process mining: a systematic literature review, International Journal of Data Science and Analytics 22 (1) (2026). doi:10.1007/s41060-025-01000-w.

[18] L. Madeyski, B. Kitchenham, M. Shepperd, LLM4SCREENLIT: Recommendations on assessing the performance of large language models for screening literature in systematic reviews, Information and Software Technology 198 (2026) 108204. doi:10.1016/j.infsof.2026.108204.
URL https://doi.org/10.1016/j.infsof.2026.108204

[19] K. Petersen, S. Vakkalanka, L. Kuzniarz, Guidelines for conducting systematic mapping studies in software engineering: An update, Information and Software Technology 64 (2015) 1–18. doi:10.1016/j.infsof.2015.03.007.

[20] B. A. Kitchenham, S. Charters, Guidelines for performing systematic literature reviews in software engineering, Tech. rep., EBSE Technical Report EBSE-2007-01 (2007).

[21] C. Wohlin, Guidelines for snowballing in systematic literature studies and a replication in software engineering, in: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE 2014), ACM, 2014. doi:10.1145/2601248.2601268.

[22] E. Syriani, I. Dávid, G. Kumar, Assessing the ability of ChatGPT to screen articles for systematic reviews, in: arXiv preprint, 2023, arXiv:2307.06464. doi:10.48550/arXiv.2307.06464.

[23] Q. Khraisha, S. Put, J. Kappenberg, A. Warraitch, K. Hadfield, Can large language models replace humans in systematic reviews? evaluating GPT-4's efficacy in screening and extracting data from peer-reviewed and grey literature in multiple languages, Research Synthesis Methods 15 (4) (2024) 616–626. doi:10.1002/jrsm.1715.

[24] J. R. Landis, G. G. Koch, The measurement of observer agreement for categorical data, Biometrics 33 (1) (1977) 159–174.

[25] T. Dybå, T. Dingsøyr, Empirical studies of agile software development: A systematic review, Information and Software Technology 50 (9–10) (2008) 833–859. doi:10.1016/j.infsof.2008.01.006.

[26] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, D. Moher, The PRISMA 2020 statement: An updated guideline for reporting systematic reviews, BMJ 372 (2021) n71. doi:10.1136/bmj.n71.

[27] S. Jo, R. Kwon, G. Kwon, Exploring collaboration patterns in GitHub using discrete time Markov chain, in: 2023 30th Asia-Pacific Software Engineering Conference (APSEC), IEEE, 2023, pp. 645–646. doi:10.1109/APSEC60848.2023.00090.

[28] S. Jo, R. Kwon, G. Kwon, Probabilistic model checking GitHub repositories for software project analysis, Applied Sciences 14 (3) (2024) 1260. doi:10.3390/app14031260.

[29] M. Ortu, G. Destefanis, T. Hall, D. Bowes, Fault-insertion and fault-fixing behavioural patterns in Apache software foundation projects, Information and Software Technology 158 (2023) 107187. doi:10.1016/j.infsof.2023.107187.

[30] R. Joshi, N. Kahani, Comparative study of reinforcement learning in GitHub pull request outcome predictions (2024) 489–500doi:10.1109/SANER60148.2024.00057.

[31] C.-K. Jeon, N. Kim, H. P. In, Probabilistic approach to predicting risk in software projects using software repository data, International Journal of Software Engineering and Knowledge Engineering 25 (6) (2015) 1017–1032. doi:10.1142/S0218194015500151.

[32] L. Tian, A. Noore, Modeling distributed software defect removal effectiveness in the presence of code churn, Mathematical and Computer Modelling 41 (4–5) (2005) 379–389. doi:10.1016/j.mcm.2004.10.021.

[33] M. Tyagi, D. K. Tyagi, L. K. Tyagi, N. Tyagi, D. Kumar, A. Sikandar, Agile planning and tracking of software projects using the state-space approach (2021) 355–371doi:10.1007/978-981-16-0404-1_26.

[34] H. Washizaki, K. Honda, Y. Fukazawa, Predicting release time for open source software based on the generalized software reliability model (2015) 76–81doi:10.1109/Agile.2015.19.

[35] I. Massitela, J. Assunção, A. R. Santos, P. Fernandes, A structured stochastic model for software project estimation in Waterfall models (2018). doi:10.18293/SEKE2018-033.

[36] W. B. Mesmia, M. Escheikh, K. Barkaoui, DevOps workflow verification and duration prediction using non-Markovian stochastic Petri nets, Journal of Software: Evolution and Process 33 (3) (2021) e2329. doi:10.1002/smr.2329.

[37] S. Bhadra, A stochastic Petri net model of continuous integration and continuous delivery, in: 2022 IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW), IEEE, 2022, pp. 114–117. doi:10.1109/ISSREW55968.2022.00050.

[38] S. Bhadra, B. Das, P. Silva, V. Nagaraju, L. Fiondella, A stochastic Petri net model of continuous integration and continuous delivery, in: 2023 Annual Reliability and Maintainability Symposium (RAMS), IEEE, 2023, pp. 1–6. doi:10.1109/RAMS51473.2023.10088212.

[39] Y. Li, H. Zhang, L. Dong, B. Liu, J. Ma, Constructing a hybrid software process simulation model in practice: An exemplar from industry, in: 2020 IEEE/ACM International Conference on Software and System Processes (ICSSP), IEEE, 2020, pp. 135–144. doi:10.1145/3379177.3388906.

[40] M. I. Lunesu, R. Tonelli, L. Marchesi, M. Marchesi, Assessing the risk of software development in agile methodologies using simulation, IEEE Access 9 (2021) 134240–134258. doi:10.1109/ACCESS.2021.3115941.

[41] T. Magennis, The economic impact of software development process choice: Cycle-time analysis and Monte Carlo simulation results, in: 2015 48th Hawaii International Conference on System Sciences (HICSS), IEEE, 2015, pp. 5055–5064. doi:10.1109/HICSS.2015.599.

[42] M. A. Guinea-Cabrera, J. A. Holgado-Terriza, P. A. Pico-Valencia, PRIMAD: Building digital twins for software development lifecycle within an evolutionary architecture, in: 2025 IEEE Smart World Congress (SWC), IEEE, 2025, pp. 1680–1685. doi:10.1109/SWC65939.2025.00261.

[43] M. Nafreen, M. Luperon, L. Fiondella, V. Nagaraju, Y. Shi, T. Wandji, Connecting software reliability growth models to software defect tracking, in: 2020 IEEE 31st International Symposium on Software Reliability Engineering (ISSRE), IEEE, 2020, pp. 138–147. doi:10.1109/ISSRE5003.2020.00022.

[44] E. Incerto, A. Vandin, S. S. Ahrabi, Stochastic conformance checking based on variable-length Markov chains, Information Systems 133 (2025) 102561. doi:10.1016/j.is.2025.102561.

[45] O. López-Pintado, M. Dumas, Discovery and simulation of business processes with probabilistic resource availability calendars, in: 2023 5th International Conference on Process Mining (ICPM), IEEE, 2023, pp. 1–8. doi:10.1109/ICPM60904.2023.10271965.

[46] P. Jalote, D. Kamma, Studying task processes for improving programmer productivity, IEEE Transactions on Software Engineering 47 (4) (2021) 801–817. doi:10.1109/TSE.2019.2904230.

[47] M. Gupta, Improving software maintenance using process mining and predictive analytics, in: 2017 IEEE International Conference on Software Maintenance and Evolution (ICSME), IEEE, 2017, pp. 681–686. doi:10.1109/ICSME.2017.39.

[48] M. Pourbafrani, S. Jiao, W. M. P. van der Aalst, SIMPT: Process improvement using interactive simulation of time-aware process trees, in: Research Challenges in Information Science (RCIS 2021), Vol. 415 of LNBIP, Springer, 2021, pp. 588–594. doi:10.1007/978-3-030-75018-3_40.

[49] A. Buliga, F. Meneghello, R. Graziosi, M. Ronzani, Enhanced what-if scenarios generation by bridging generative models and process simulation, in: 2025 7th International Conference on Process Mining (ICPM), IEEE, 2025, pp. 1–8. doi:10.1109/ICPM66919.2025.11220730.

[50] J. Caldeira, F. B. e Abreu, J. Cardoso, J. Reis, Unveiling process insights from refactoring practices, Computer Standards & Interfaces 81 (2022) 103587. doi:10.1016/j.csi.2021.103587.